CONTEXT FREE GRAMMARS AND LANGUAGES

A Grammar G is 4-tuple or quadruple $G = (V, T, P, S)$ where

V is set of variables or non-terminals

T is set of terminals

P is set of productions.

Each production is of the form $\alpha \rightarrow \beta$ where α is a string in $(VUT)^+$ and hence α can not be ε and ε can not occur on the left hand side of any production. But, β is a string in $(VUT)^*$ and hence it includes ε also. So, ε can occur on the right hand side of the production.

S is the start symbol.

Examples :

Non-Terminals or variables (V)

- The lower case names such as expression, operator, operand, statement etc.,
- The capital letters near the beginning of the alphabets such as A, B, C, D, etc.,
- The letter S is the start symbol.

Terminals (T)

- The keywords such as if, for, while, do-while etc.,
- Digits from 0 to 9
- Symbols such as +, -, *, /, etc.,
- The lower case letters near the beginning of the alphabets such as a, b, c, d etc.,
- The bold faced letters such as id.
- The lower case letters near the end of the alphabets such as u, v, w, x, y, z represent string of terminals.

Grammar Symbols

The grammar symbol can be terminal or non-terminal.

- Greek letters
- The capital letters near the end of the alphabets such as X, Y, Z, etc.,

Chomsky Hierarchy

Noam Chomsky who is the founder of formal language theory has classified the grammar into various categories.

Types of Grammars → Type 0 grammar (Phrase structured grammar)
→ Type 1 grammar (Context sensitive grammar)
→ Type 2 grammar (Context Free grammar)
→ Type 3 grammar (Regular grammar)

* Type 0 Grammar

A grammar $G = (V, T, P, S)$ is said to be type 0 grammar or unrestricted or Phrase structured grammar if all the productions are of the form $\alpha \to \beta$ where

$\alpha \in (V \cup T)^+$ and $\beta \in (V \cup T)^*$ $\quad X \in (V \cup \Sigma)^+$ & $Y \in (V \cup \Sigma)^*$

In this type of grammar there are no restrictions on length of α and β. The only restriction is that α can not be ε. i.e., ε can not appear on the left hand side of any production. But, ε can appear on the right hand side of the production.

This is the largest family of grammars more powerful than all other types of grammars. Any language can be obtained from this grammar.

The language generated from this grammar is called type 0 language or recursively enumerable language. Only Turing machine can recognize this language.

Example : $S \to aAb \mid \varepsilon$
$aA \to bAA$
$bA \to a$

* Type 1 Grammar

A grammar $G=(V,T,P,S)$ is said to be type1 grammar or context sensitive grammar if all the productions are of the form $\alpha \to \beta$. But, there is a restriction on length of β. The length of β must be at least as much as the length of α i.e, $|\beta| \geq |\alpha|$ and α & $\beta \in (V \cup T)^+$ i.e., ε can not appear on the left hand or right hand side of any production. It is an ε-free grammar.

$X \to Y \quad |Y| \geq |X| \quad X \& Y \in (V \cup \Sigma)^+$

The language generated from this grammar is called type1 language or context sensitive language.

Linear Bounded Automata (LBA) can be constructed to recognize the language generated from this grammar.

Example : $S \to aAb$

$aA \to bAA$

$bA \to aa$

* Type 2 Grammar

A grammar $G=(V,T,P,S)$ is said to be type2 grammar or context free grammar if all the productions are of the form $A \to \alpha$ where $\alpha \in (V \cup T)^*$ and A is non-terminal. The symbol ε can appear on the right hand side of any production. $X \to Y \quad Y \in (V \cup \Sigma)^*$

The language generated from this grammar is called type 2 language or context free language.

Pushdown Automaton (PDA) can be constructed to recognize the language generated from this grammar.

Example : $S \to aB \mid bA \mid \varepsilon$

$A \to aA \mid b$

$B \to bB \mid a \mid \varepsilon$

$A \to X \mid X \in (V \cup \Sigma)^*$

↓ Non-terminal

* Type 3 Grammar or Regular Grammar

The grammar $G = (V, T, P, S)$ is said to be type 3 grammar or regular grammar iff the grammar is right linear or left linear.

A grammar G is said to be right linear if all the productions are of the form

$$A \rightarrow \omega B$$

or

$$A \rightarrow \omega$$

where A, B are variables and $\omega \in T^*$ i.e., ω is string of terminals.

A grammar G is said to be left linear if all the productions are of the form

$$A \rightarrow B\omega$$

or

$$A \rightarrow \omega$$

where $A, B \in V$ and $\omega \in T^*$.

Example for right linear grammar

$S \rightarrow aaB \mid bbA \mid \varepsilon$

$A \rightarrow aA \mid b$

$B \rightarrow bB \mid a \mid \varepsilon$

NOTE: ε and string of terminals can appear on RHS of any production and if non-terminal is present on RHS of any production, only one non-terminal should be present and it has to be the right most symbol on RHS.

Example for left linear grammar

$S \rightarrow Baa \mid Abb \mid \varepsilon$

$A \rightarrow Aa \mid b$

$B \rightarrow Bb \mid a \mid \varepsilon$

NOTE: ε and string of terminals can appear on RHS of any production and if non-terminal is present on RHS of any production, only one non-terminal should be present and it has to be the left most symbol on RHS.

NOTE: Consider the following grammar

$S \rightarrow aA$

$A \rightarrow aB \mid b$

$B \rightarrow Ab \mid a$

In this grammar, each production is either left linear or right linear. But, the grammar is not either left linear or right linear. Such type of grammar is called linear grammar. So, a grammar which has at most one non-terminal on the right side of any production without restriction on the position of this non-terminal is called linear grammar.

The language generated from the regular grammar is called regular language. The language accepted by FA is regular language. So, we can construct a finite automaton given a regular grammar and vice versa.

Grammar from finite Automata

Procedure: Let $M = (Q, \Sigma, \delta, q_0, F)$ be a finite automata accepting L where $Q = \{q_0, q_1, \ldots, q_n\}$

$\Sigma = \{a_1, a_2, \ldots, a_m\}$

A grammar $G = (V, T, P, S)$ can be constructed where

- $V = \{q_0, q_1, \ldots, q_n\}$ i.e., the states of DFA will be the variables in the grammar.
- $T = \Sigma$ i.e., the input alphabets of DFA are the terminals in grammar.
- $S = q_0$ i.e., the start state of DFA is the start symbol in the grammar.
- The productions P from the transitions can be obtained as shown below:

STEP 1 : If $\delta(q_i, a) = q_j$ then introduce the ~~transition~~ production

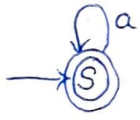
$q_i \rightarrow a q_j$

STEP 2 : If $q \in F$ i.e., if q is the final state in FA, then introduce the production

$q \rightarrow \varepsilon$

Q1. Obtain grammar to generate string consisting of any number of a's.

$\Sigma = \{a\}$



$\delta(S, a) = S \qquad S \rightarrow aS$

S is a final state $\quad S \rightarrow \varepsilon$

$G = (V, T, P, S)$

$V = \{S\}$

$T = \{a\}$

$P = \{ S \rightarrow aS, S \rightarrow \varepsilon \}$ S is the start state

The grammar to generate string consisting of any number of a's is given by

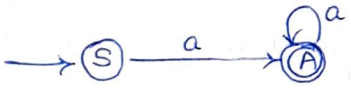
$S \rightarrow aS$

$S \rightarrow \varepsilon$

$G = (V, \Sigma, R, S)$

Q2. Obtain grammar to generate string consisting of at least one a.

$\Sigma = \{a\}$



$\delta(S, a) = A$

$\delta(A, a) = A$

$S \rightarrow aA$

$A \rightarrow aA$

$A \rightarrow \varepsilon$

$S \rightarrow aA$

$A \rightarrow aA \mid \varepsilon$

(OR)

$S \rightarrow aS \mid a$

(OR)

$S \rightarrow aS \mid a$

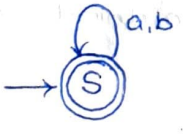
(OR)

$S \rightarrow a$

$S \rightarrow aS$

Q3. Obtain grammar to generate string consisting of any number of a's and b's.

$\Sigma = \{a, b\}$



$S \rightarrow aS$

$S \rightarrow bS$

$S \rightarrow \varepsilon$

(OR) $\quad S \rightarrow \varepsilon \mid aS \mid bS$

Q4. Obtain grammar to generate string consisting of at least two a's.

$\Sigma = \{a\}$

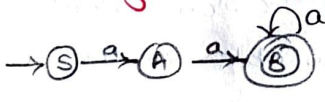
$S \rightarrow aa$

$S \rightarrow aS$ (OR)

$S \rightarrow aA$

$A \rightarrow aB$

$B \rightarrow \varepsilon \mid aB$

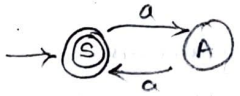


Q5. Obtain grammar to generate string consisting of even number of a's.

$\Sigma = \{a\}$

$S \rightarrow \varepsilon$

$S \rightarrow aaS$



$S \rightarrow aA \mid \varepsilon$

$A \rightarrow aS$

(OR)

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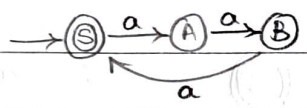
Q6. Obtain grammar to generate string consisting of multiples of three a's.

$S \to aaaS$ (OR) $S \to \varepsilon \mid aaaS$ 0, 3, 6, 9 . . .

$S \to \varepsilon$

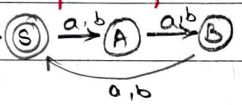


$S \to aA \mid \varepsilon$
$A \to aB$
$B \to aS$

Q7. Obtain grammar to generate strings of a's and b's such that string length is multiple of 3.

$S \to \varepsilon \mid AAAS$

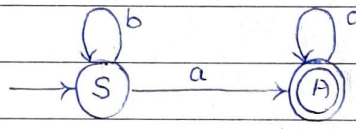
$A \to a \mid b$



$S \to aA \mid bA \mid \varepsilon$ 0, 3, 6, 9 . . . .
$A \to aB \mid bB$
$B \to aS \mid bS$

Q8. Obtain grammar to generate string consisting of any number of a's and b's with at least one a.

$S \to a$
$S \to aS$ (OR)
$S \to bS$
$S \to a \mid aS \mid bS$



same

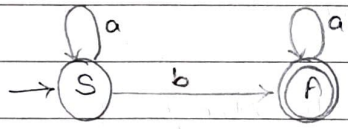
$S \to bS$
$S \to aA$
$A \to aA$
(OR)
$A \to bA$
$A \to \varepsilon$

$S \to aA \mid bS$
$A \to aA \mid bA \mid \varepsilon$

Q9. Obtain grammar to generate string consisting of any number of a's and b's with at least one b.

$S \to b$
$S \to aS$
$S \to bS$

(OR)

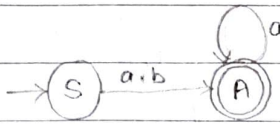


$S \to aS \mid bA$
$A \to aA \mid bA \mid \varepsilon$

Q10. Obtain grammar to generate string consisting of any number of a's and b's with at least one a or at least one b.

$S \to a \mid b$
$S \to aS$
$S \to bS$

(OR)



$S \to aA \mid bA$
$A \to aA \mid bA \mid \varepsilon$

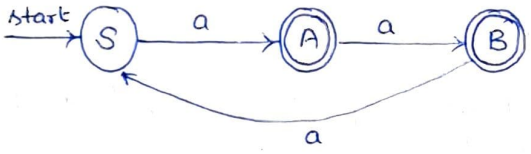
Q11. Obtain grammar to accept the following language

$L = \{ \omega : |\omega| \bmod 3 > 0 \text{ where } \omega \in \{a\}^* \}$ ∅, 1, 2

$S \rightarrow a \mid aa$

$S \rightarrow aaaS$

(OR)



$S \rightarrow aA$

$A \rightarrow aB \mid \varepsilon$

$B \rightarrow aS \mid \varepsilon$

Grammar from Regular Expressions

Q1. Obtain grammar to generate strings of a's and b's having a substring ab.

$(a+b)^*\ ab\ (a+b)^*$

$S \rightarrow AabA$

$A \rightarrow \varepsilon \mid aA \mid bA$

Q2. Obtain grammar to generate strings of a's and b's ending with string ab.

$(a+b)^*\ ab$

$S \rightarrow Aab$

$A \rightarrow \varepsilon \mid aA \mid bA$

Q3. Obtain grammar to generate strings of a's and b's starting with ab.

$ab\ (a+b)^*$

$S \rightarrow abA$

$A \rightarrow \varepsilon \mid aA \mid bA$

Q4. Obtain grammar to generate the following language

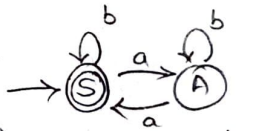
$L = \{ \omega : n_a(\omega) \bmod 2 = 0 \text{ where } \omega \in \{a,b\}^* \}$

$S \rightarrow \varepsilon \mid bS \mid SaSaS$

$^*((\varepsilon+b)\ a\ (\varepsilon+b)\ a\ (\varepsilon+b))^*$

0, 2, 4, 6, 8, ... (a's)

Any no of b's



$S \rightarrow bS \mid aA \mid \varepsilon$

$A \rightarrow aS \mid bA$

Derivation

Let $A \to \alpha B \gamma$ and $B \to \beta$ are productions in grammar G, where α, β and γ are strings of terminals and/or non-terminals, A and B are non-terminals. The non-terminal A produces the string $\alpha\beta\gamma$ by replacing the non-terminal B in $\alpha B \gamma$ by the string β by applying the production $B \to \beta$ and can be written as

$$A \Rightarrow \alpha\beta\gamma$$

The process of obtaining string of terminals and/or non-terminals from the start symbol by applying some or all productions is called derivation.

If a string is obtained by applying only one production, then it is called one-step derivation and is denoted by the symbol '$\Rightarrow$'

If one or more productions are applied to get the string $\alpha\beta\gamma$ from A, then

$$A \overset{+}{\Rightarrow} \alpha\beta\gamma$$

If zero or more productions are applied to get the string $\alpha\beta\gamma$ from A, then $A \overset{*}{\Rightarrow} \alpha\beta\gamma$

Q1. Consider the grammar shown below from which any arithmetic expression can be obtained

$E \to E + E$

$E \to E - E$

$E \to E * E$

$E \to E / E$

$E \to id$

Obtain the string id + id * id and show the derivation for the same.

$E \Rightarrow E + E$ Leftmost Derivation

$\Rightarrow id + E$

$\Rightarrow id + E * E \Rightarrow id + id * E \Rightarrow id + id * id$

Since the string id + id * id is obtained from the start symbol E by applying more than one production, this can be written as

$$E \Rightarrow^{+} id + id * id$$

Sentence

Let $G = (V, T, P, S)$ be a grammar. The string ω obtained from the grammar G such that $S \Rightarrow^{+} \omega$ is called sentence of grammar G. Here, ω is the string of terminals.

If there is a derivation $S \Rightarrow^{*} \alpha$, where α contains string of terminals and/or non-terminals, then α is called sentential form of G.

Example: id + id * id is the sentence of the grammar.

E + E, id + E, id + E * E, id + id * E, id + id * id are all sentential forms of the grammar.

Language

A given grammar can generate a set of strings consisting of only of terminals by applying productions in different order. The set of such strings is called the language.

Let $G = (V, T, P, S)$ be a grammar. The language $L(G)$ generated by the grammar G is

$$L(G) = \{\omega \mid S \Rightarrow^{*} \omega \text{ and } \omega \in T^{*}\}$$

i.e., ω is a string of terminals (may be ε) obtained from the start symbol S by applying an arbitrary number of productions.

NOTE: The intermediate string of terminals and/or non-terminals obtained during the derivation process is called sentential form of G. The various strings which are the elements of $L(G)$ are called sentences.

Q1. Consider the following grammar $S \rightarrow aCa$

$C \rightarrow aCa \mid b$

What is the language generated by this grammar?

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$S \Rightarrow aCa \Rightarrow aba \in L(G)$

$S \Rightarrow aCa$
$\Rightarrow aaCaa$
$\Rightarrow aaaCaaa$
$\Rightarrow a^nCa^n$
$\Rightarrow a^nba^n$

$L(G) = \{a^nba^n \mid n \geq 1\}$

Q1. Consider the following grammar

$S \rightarrow aCa$

$C \rightarrow aCa \mid b$

What is the language generated by this grammar?

Grammars for Other Languages

Q1. Obtain a grammar to generate the following language.

$L: \{a^nb^n : n \geq 0\}$

$S \rightarrow \varepsilon \mid aSb$

0 ε
1 ab
2 aabb
3 aaabbb

Q2. Obtain a grammar to generate the following language

$L: \{a^nb^n : n \geq 1\}$

$S \rightarrow ab \mid aSb$

1 ab
2 aabb
3 aaabbb

Q3. Obtain a grammar to generate the following language

$L: \{a^{n+1}b^n : n \geq 0\}$

$S \rightarrow a \mid aSb$

0 a
1 aab
2 aaabb

Q4. Obtain a grammar to generate the following language

$L: \{a^nb^{n+1} : n \geq 0\}$

$S \rightarrow b \mid aSb$

Q5. Obtain a grammar to generate the following language

$L: \{a^nb^{n+2} : n \geq 0\}$

$S \rightarrow bb \mid aSb$

0 bb
1 abbb

Q6. Obtain a grammar to generate the following language

$L = \{a^n b^{2n} : n \geq 0\}$

$S \to \varepsilon \mid aSbb$

0	ε
1	abb
2	aabbbb

$L = \{a^n b^{2n} : n \geq 1\}$

$S \to abb \mid aSbb$

Q7. Let $\Sigma = \{a, b\}$. Obtain a grammar G generating set of all palindromes over Σ.

The grammar to generate strings of palindromes over Σ is given by $G = (V, T, P, S)$ where

$V = \{S\}$

$T = \{a, b\}$

S is the start symbol

$P = \{ \quad S \to \varepsilon$

$S \to a \mid b$

$S \to aSa \mid bSb$

$\}$

ε

a b

aa bb

ω - awa bwb

Q8. Obtain a grammar to generate the following language

$L = \{ \omega\omega^R$ where $\omega \in \{a, b\}^* \}$

$S \to \varepsilon$

$S \to aSa \mid bSb$

abab baba

ω ω^R

$S \to aSa \mid bSb \mid \varepsilon$

NOTE : $L = \{\omega c \omega^R \mid \omega \in \{a, b\}^*\}$

$S \to c$

$S \to aSa \mid bSb$

Q9. Obtain a grammar to generate a language consisting of all non-palindromes over $\{a, b\}$

The grammar to generate strings of non-palindromes is given by $G = (V, T, P, S)$ where

$V = \{S, A, B\}$

$T = \{a, b\}$

S is the start symbol

$P = \{ \quad S \to aSa \mid bSb$ abba | aa | bb

$S \to A$

$A \to aBb \mid bBa$ aabb | ab | ba abab

$B \to aB \mid bB \mid \varepsilon$

$\}$

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Q10. Obtain the grammar to generate the language

$L: \{ 0^m 1^m 0^n \mid m \geq 1 \text{ and } n \geq 0 \}$

$S \rightarrow AB$

$A \rightarrow 01 \mid 0A1$

$B \rightarrow \varepsilon \mid 2B$

(OR)

$S \rightarrow A \mid S2$

$A \rightarrow 01 \mid 0A1$

Q11. Obtain the grammar to generate the language

$L: \{ \omega \mid n_a(\omega) = n_b(\omega) \}$

The grammar to generate the language $L: \{ \omega \mid n_a(\omega) = n_b(\omega) \}$ is

$G = (V, T, P, S)$ where

$V = \{S\}$

$T = \{a, b\}$

S is the start symbol

P = {

$S \rightarrow \varepsilon \mid aSb \mid bSa \mid SS$

}

ε
ab
ba
aabb
baba
abab
bbaa
{abba SS}
aaabbb + ababab abbaab
bbbaaa +

Q12. What is the language generated by the grammar

$S \rightarrow 0A \mid \varepsilon$

$A \rightarrow 1S$

$S \rightarrow \varepsilon$

$S \Rightarrow 0A$

$\Rightarrow 01S$

$\Rightarrow 010A$

$\Rightarrow 0101S$

$\Rightarrow 0101$

$L: \{(01)^n \mid n \geq 0\}$

NOTE : 1) $S \rightarrow \varepsilon \mid 01S$

2) $S \rightarrow \varepsilon \mid S01$

Q13. Obtain a CFG to generate a string of balanced parentheses

The grammar G to generate a string of balanced parentheses is given by $G = (V, T, P, S)$ where

$V = \{S\}$

$T = \{(,)\}$

S is the start symbol

$P = \{ \quad S \to \varepsilon \mid (S) \mid SS$

$\}$

$A \to \alpha$

(())

~~)()~~ $\to SS$

Q14. Obtain a grammar to generate the Language

$L = \{0^i 1^j \mid i \neq j, i \geq 0 \text{ and } j \geq 0\}$

$G = (V, T, P, S)$

$V = \{S, A, B\}$

$T = \{0, 1\}$

S is the start symbol

$P = \{ \quad S \to 0S1$

$S \to A \mid B$

$A \to 0A \mid 0$

$B \to 1B \mid 1$

$\}$

(OR)

$G = (V, T, P, S)$

$V = \{S, A, B, C\}$

$T = \{0, 1\}$

S is the start symbol

$P = \{ \quad S \to BA$ At least one 0

$S \to AC$ " " 1

$A \to 0A1 \mid \varepsilon$

$B \to 0B \mid 0$

$C \to 1C \mid 1$

$\}$

Q15. Obtain a grammar to generate the language

$L = \{a^{n+2} b^m \mid n \geq 0 \text{ and } m > n\}$

$S \to aAB$

$A \to aAb \mid ab$

$B \to bB \mid \varepsilon$

$aab b^*$

$aaabbb^*$

$aaaabbbb^*$

$a^n b^n \mid n \geq 1$

aAB

$n = 0 \quad m > n \quad 1 > 0$

$n = 1 \quad m > n \quad 2 > 1$

Q16. Obtain a grammar to generate the Language

$L = \{a^n b^m \mid n \geq 0, m > n\}$

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$a^n b^m$
$n \geq 0$,
$m > n$

$G = (V, T, P, S)$ $\quad n=0 \quad n=1 \quad n=2$

$V = \{S, A, B\}$ $\quad m \geq 1 \quad m \geq 2 \quad m \geq 3$

$T = \{a, b\}$ $\quad \varepsilon bb^*, \ ab\,bb^*, \ aabbbb^* \ldots$

S is the start symbol $\quad a^n b^n \mid n \geq 0$

P: {

$S \rightarrow AB$ (OR) $S \rightarrow aSb \mid B$

$A \rightarrow aAb \mid \varepsilon$ $\quad B \rightarrow bB \mid b$

$B \rightarrow bB \mid b$

}

Q17. Obtain a grammar to generate the language

$L = \{ a^n b^{n-3} \mid n \geq 3 \}$

$G = (V, T, P, S)$ $\quad n=3 \quad n=4 \quad n=5$

$V = \{S, A\}$ $\quad aaa \quad aaaab \quad aaaabb \ldots$

$T = \{a, b\}$ $\quad a^n b^n \mid n \geq 0$

S is the start symbol

P: { $S \rightarrow aaaA$ (OR) $S \rightarrow aSb \mid aaa$

$A \rightarrow aAb \mid \varepsilon$

}

Q18. Obtain a grammar to generate the language

$L = L_1 L_2$ where $L_1 = \{ a^n b^m \mid n \geq 0, m > n \}$

$L_2 = \{ 0^n 1^{2n} \mid n \geq 0 \}$

L_1 $\quad L_2$

$S_1 \rightarrow aS_1b \mid B$ $\quad S_2 \rightarrow 0S_2 11 \mid \varepsilon$

$B \rightarrow bB \mid b$

$V = \{ S_1, S_2, S, B \}$ $\quad T = \{a, b, 0, 1\}$ $\quad S$ is the start symbol.

P: { $S \rightarrow S_1 S_2$

$S_1 \rightarrow aS_1b \mid B$ (OR) $S \rightarrow S_1 S_2$

$S_2 \rightarrow 0S_2 11 \mid \varepsilon$ $\quad S_1 \rightarrow AB$ $\quad A \rightarrow aAb \mid \varepsilon$

$B \rightarrow bB \mid b$ $\quad S_2 \rightarrow 0S_2 11 \mid \varepsilon$ $\quad B \rightarrow bB \mid b$

}

Q19. Obtain a grammar to generate the language $L = L_1 \cup L_2$ where

$L_1 = \{a^n b^m \mid n \geq 0, m > n\}$

$L_2 = \{0^{n+1} 1^{2n} \mid n \geq 0\}$

$G = (V, T, P, S)$

$V = \{S, S_1, S_2, A, B\}$

$T = \{a, b, 0, 1\}$

S is the start symbol

P : { $S \to S_1 \mid S_2$

$S_1 \to AB$

$S_2 \to 0S_2 11 \mid \varepsilon$

$A \to aAb \mid \varepsilon$

$B \to bB \mid b$ }

(OR)

$G = (V, T, P, S)$

$V = \{S, S_1, S_2, B\}$

$T = \{a, b, 0, 1\}$

S is the start symbol

P : { $S \to S_1 \mid S_2$

$S_1 \to aS_1 b \mid aBb$

$S_2 \to 0S_2 11 \mid \varepsilon$

$B \to bB \mid b$ }

Q20. Obtain a grammar to generate the language

$L = \{\omega : |\omega| \bmod 3 \neq |\omega| \bmod 2\}$ on $\Sigma = \{a\}$

$S \to aa \mid aaa \mid aaaa \mid aaaaa \mid aaaaaa S$

(2) (3) (4) (5) (6) (5)

$\{0, 1, 2\}$ $\{0, 1\}$ $(\varepsilon \& a)$ ×

0, 1, 7, 12, 17, 22, 27, 32 ✓

Q21. Obtain a grammar to generate the language

$L = \{\omega : |\omega| \bmod 3 \geq |\omega| \bmod 2\}$ on $\Sigma = \{a\}$

$S \to \varepsilon \mid a \mid aa \mid aaaa \mid aaaaa \mid aaaaaa S$

(1) (2) (4) (5) (6)

(0,1,2) (0,1)

$0 \geq 1$ $1 \geq 0$ (4)

$1 \geq 1$ $2 \geq 1$ (5)

$2 \geq 0, 1$ $0 \geq 0$ (6)

$0 \geq 3$ ✗ $1 \geq 1$ (7)

$2 \geq 0$ (8)

$0 \geq 1$ (9) ×

(10)

$0 \geq 1$ (15)

$0 \geq 0$ (20) ✓

$0 \geq 0$ ε

$1 \geq 0$ a

$2 \geq 0$ aa

$0 \geq 1$ ×

$1 \geq 1$ a

$2 \geq 1$ aa

3 ×, 9 ×

aS × (3) $aaaaS$ × (9) $aaaaaaS$

aaS × (3) $aaaaaS$ × (15)

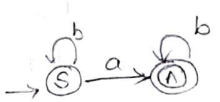
aaa^x

a^9 ×

Q22. Obtain a grammar to generate set of all strings with exactly one a when $\Sigma = \{a, b\}$

$S \to aB \mid bS$

$B \to bB \mid \varepsilon$



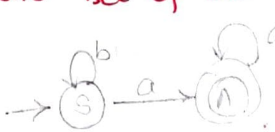
$S \to bS \mid aA$

$A \to \varepsilon \mid bA$

Q23. Obtain a grammar to generate set of all strings with at least one a when $\Sigma = \{a, b\}$

$S \to bS \mid aA$

$A \to aA \mid bA \mid \varepsilon$



Q24. Obtain a grammar to generate the set of all strings with no more than three a's when $\Sigma = \{a, b\}$

$S \to bS \mid aA \mid \varepsilon$

$A \to bA \mid aB \mid \varepsilon$

$B \to bB \mid aC \mid \varepsilon$

$C \to bC \mid \varepsilon$

(OR)

$S \to B \mid BaB \mid BaBaB \mid BaBaBaB$

$B \to bB \mid \varepsilon$

Q25. Obtain a grammar to generate the language $L = \{ w \mid n_a(w) = n_b(w) \}$

(NOTE)

$S \to AaA$

$A \to aAb \mid bAa \mid AA \mid \varepsilon$

$n_a(w) + 1 = n_b(w)$ (+1)

(OR)

$S \to AbA$

$A \to aAb \mid bAa \mid AA \mid \varepsilon$

$n_a(w) = n_b(w) + 1$

Q26. Obtain the grammar to generate the language

$L = \{ w \mid n_a(w) > n_b(w) \}$

$S \to AB \mid BA \mid ABA$

$A \to aAb$

$A \to bAa$

$A \to AA$

$A \to \varepsilon$

$B \to aB \mid a$

$A \to aAb \mid bAa \mid AA \mid \varepsilon$

(OR)

$S \to a \mid aSb \mid bSa \mid SS$

Q27. Obtain a grammar to generate a language of strings of 0's and 1's having a substring 000.

$S \to A000A$

$A \to 0A \mid 1A \mid \varepsilon$

$(001)^* 000 (001)^*$

Q28. Obtain a grammar to generate the following language

$L = \{ a^n b^m c^k \mid n+2m = k \text{ for } n \geq 0, m \geq 0 \}$

$S \to aSc \mid A$

$A \to bAcc \mid \varepsilon$

$a^n b^m c^{n+2m} \Rightarrow a^n b^m c^{2m} c^n \Rightarrow a^n c^n b^m c^{2m}$

Q29. Obtain a grammar to generate integers.

$G = (V, T, P, S)$

$V = \{D, S, N, I\}$

$T = \{+, -, 0, 1, 2, \ldots, 9\}$

S: I is the start symbol

P: {

$I \rightarrow N \mid SN$

$N \rightarrow D \mid ND \mid DN$

$S \rightarrow + \mid - \mid \varepsilon$

$D \rightarrow 0 \mid 1 \mid 2 \mid \ldots \mid 9$

}

Q30. Let $G = (V, T, P, S)$ be a CFG where $V = \{S\}$

$T = \{a, b\}$

$P = \{ S \rightarrow aSa \mid bSb \mid \varepsilon$

S is the start symbol.

$A \rightarrow \alpha$

What is the language generated by this grammar?

$L = \{ ww^R \mid w \in \{a+b\}^* \}$

Q31. Obtain a grammar to generate an arithmetic expression using the operators +, −, *, / and ^. An identifier can start with any of the letters from {a, b, c} and can be followed by zero or more symbols from {a, b, c}.

$G = (V, T, P, S)$

$V = \{E, I\}$

$T = \{+, -, *, /, ^, a, b, c\}$

S is the start symbol

P = {

$E \rightarrow I$

$E \rightarrow E+E \mid E-E \mid E*E \mid E/E \mid E^E \mid (E)$

$I \rightarrow Ia \mid Ib \mid Ic \mid a \mid b \mid c$

}

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Leftmost Derivation

In the derivation process if a left most variable is replaced at every step, then the derivation is said to be leftmost.

Example:

Grammar	Leftmost derivation for the string id + id * id
$E \to E+E$	
$E \to E-E$	$E \underset{lm}{\Rightarrow} E+E$
$E \to E*E$	$\underset{lm}{\Rightarrow} id+E$
$E \to E/E$	$\underset{lm}{\Rightarrow} id+E*E$ (OR) $E \overset{+}{\underset{lm}{\Rightarrow}} id+id*id$
$E \to id$	$\underset{lm}{\Rightarrow} id+id*E$
	$\underset{lm}{\Rightarrow} id+id*id$

Rightmost Derivation

In the derivation process if a right most variable is replaced at every step, then the derivation is said to be rightmost.

Example:

Grammar	Rightmost derivation for the string id + id * id
$E \to E+E$	
$E \to E-E$	$E \underset{rm}{\Rightarrow} E+E$
$E \to E*E$	$\underset{rm}{\Rightarrow} E+E*E$
$E \to E/E$	$\underset{rm}{\Rightarrow} E+E*id$ (OR) $E \overset{+}{\underset{rm}{\Rightarrow}} id+id*id$
$E \to id$	$\underset{rm}{\Rightarrow} E+id*id$
	$\underset{rm}{\Rightarrow} id+id*id$

NOTE :
- Let $G=(V,T,P,S)$ be a CFG. Any string $\omega \in (V \cup T)^*$ which is derivable from the start symbol S denoted by $S \overset{*}{\Rightarrow} \alpha$ is called a sential form of G.
- If there is a derivation of the form $S \overset{*}{\underset{lm}{\Rightarrow}} \alpha$, where at each step in the derivation process only a left most variable is replaced, then α is called left-sentential form.
- If there is a derivation of the form $S \overset{*}{\underset{rm}{\Rightarrow}} \alpha$, where at each step in the derivation process only a right most variable is replaced, then α is called right-sentential form.

Q1. Obtain the leftmost derivation for the string aaabbabbba using the following grammar

$S \rightarrow aB \mid bA$

$A \rightarrow aS \mid bAA \mid a$

$B \rightarrow bS \mid aBB \mid b$

$S \underset{lm}{\Rightarrow} aB$ ($S \rightarrow aB$)

$\Rightarrow aaBB$ ($B \rightarrow aBB$)

$\Rightarrow aaaBBB$ ($B \rightarrow aBB$)

$\Rightarrow aaabBB$ ($B \rightarrow b$)

$\Rightarrow aaabbB$ ($B \rightarrow b$)

$\Rightarrow aaabbaBB$ ($B \rightarrow aBB$)

$\Rightarrow aaabbabB$ ($B \rightarrow b$)

$\Rightarrow aaabbabbS$ ($B \rightarrow bS$)

$\Rightarrow aaabbabbbA$ ($S \rightarrow bA$)

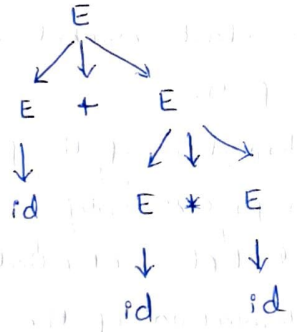
$\Rightarrow aaabbabbba$ ($A \rightarrow a$)

Derivation Tree / Parse Tree

Let $G = (V, T, P, S)$ be a CFG. The tree is derivation tree / parse tree with the following properties

1) The root has the label S.
2) Every vertex has a label which is in $(V \cup T \cup \varepsilon)$
3) Every leaf node has label from T and an interior vertex has a label from V.
4) If a vertex is labeled A and if $X_1, X_2, X_3, \ldots, X_n$ are all children of A from left, then $A \rightarrow X_1 X_2 X_3 \ldots X_n$ must be a production in P.

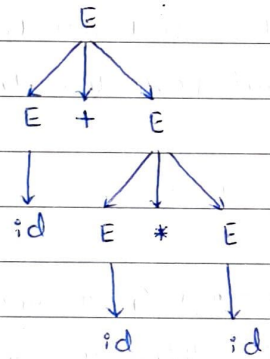
Example : The parse tree for the right most derivation for the string id+id*id



Yield of a tree

The yield of a tree is the string of symbols obtained by only reading the leaves of the tree from left to right without considering the ε-symbols. The yield of the tree is derived always from the root and the yield of the tree is always a terminal string.

Ex: Consider the parse tree



If we read only the terminal symbols in this tree from left to right we get id + id * id and is the yield of the given parse tree.

Partial Parse Tree / Partial Derivation Tree

Let $G = (V, T, P, S)$ be a CFG. The tree is partial derivation tree with the following properties:

1) The root has the label S
2) Every vertex has a label which is in $(V \cup T \cup \varepsilon)$
3) Every leaf node has label from $(V \cup T \cup \varepsilon)$
4) If a vertex is labeled A and if $X_1, X_2, X_3, \ldots, X_n$ are all children of A from left, then $A \rightarrow X_1 X_2 X_3 \ldots X_n$ must be a production in P.

Ex: Consider the partial derivation for the grammar

$E \rightarrow E + E \mid E * E \mid id$

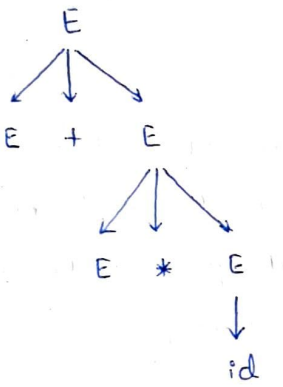
is shown below

$E \Rightarrow E + E$

$\Rightarrow E + E * E$

$\Rightarrow E + E * id$

for this partial right most derivation, the partial derivation tree is



It is clear from the parse tree and partial parse tree that all the leaves in parse tree are the symbols from $(T \cup \varepsilon)$ whereas in partial parse tree the leaves will be from $(V \cup T \cup \varepsilon)$.

Ambiguous Grammar

Let $G = (V, T, P, S)$ be a context free grammar. A grammar G is ambiguous if and only if there exists at least one string $\omega \in T^*$ for which two or more different parse trees exist by applying either the left most derivation or right most derivation.

NOTE:

* Obtain the leftmost derivation and get a string ω. Obtain the right most derivation and get a string ω. for both the derivations construct the parse tree. If there are two different parse trees, then the grammar is ambiguous.
* Obtain the string ω by applying leftmost derivation twice and construct the parse tree. If the two parse trees are different, the grammar is ambiguous.
* Obtain the string ω by applying rightmost derivation twice and construct the parse tree. If the two parse trees are different, the grammar is ambiguous.
* Apply the leftmost derivation and get a string. Apply the leftmost derivation again and get a different string. The parse trees obtained will be different and do not come to the conclusion that the grammar is ambiguous.

Q1. Consider the grammar shown below from which any arithmetic expression can be obtained

$E \rightarrow E+E$

$E \rightarrow E*E$

$E \rightarrow E-E$

$E \rightarrow E/E$

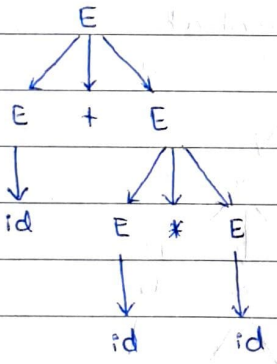
$E \rightarrow (E)/I$

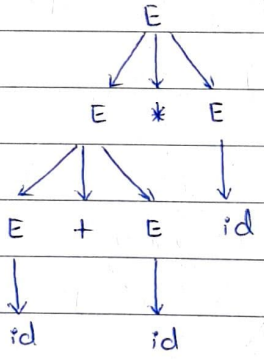
$I \rightarrow id$ Show that the grammar is ambiguous.

Sentence is id + id * id

$E \Rightarrow E+E$
$\Rightarrow id+E$
$\Rightarrow id+E*E$
$\Rightarrow id+id*E$
$\Rightarrow id+id*id$

$E \Rightarrow E*E$
$\Rightarrow E+E*E$
$\Rightarrow id+E*E$
$\Rightarrow id+id*E$
$\Rightarrow id+id*id$





Since the two parse trees are different for the same sentence id+id*id by applying leftmost derivation, the grammar is ambiguous.

Q2. Is the following grammar ambiguous?

$S \rightarrow aS \mid X$

$X \rightarrow aX \mid a$

String is aaaa

$S \Rightarrow aS$
$\Rightarrow aaS$
$\Rightarrow aaaS$
$\Rightarrow aaaX$
$\Rightarrow aaaa$

$S \Rightarrow X$
$\Rightarrow aX$
$\Rightarrow aaX$
$\Rightarrow aaaX$
$\Rightarrow aaaa$

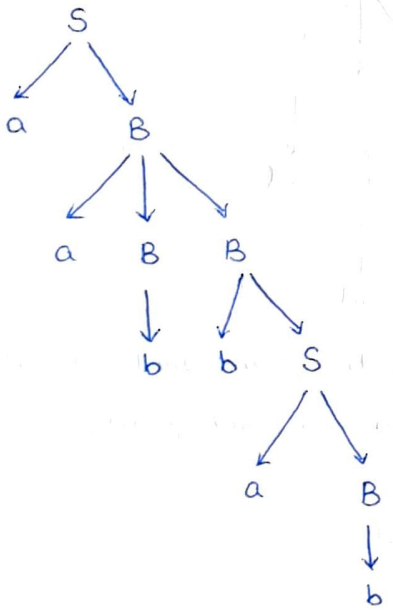
Since there are two leftmost derivations for the same sentence aaaa, the given grammar is ambiguous.

Q3. In the following grammar ambiguous?

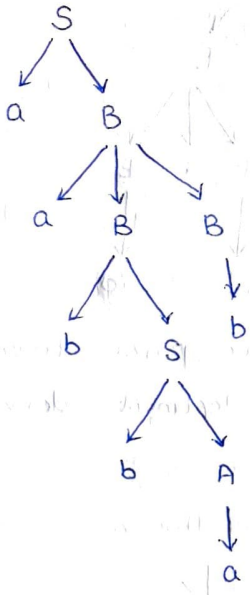
$S \rightarrow aB \mid bA$
$A \rightarrow aS \mid bAA \mid a$
$B \rightarrow bS \mid aBB \mid b$

String is aabbab

$S \Rightarrow aB$ $(S \rightarrow aB)$
$\Rightarrow aaBB$ $(B \rightarrow aBB)$
$\Rightarrow aabB$ $(B \rightarrow b)$
$\Rightarrow aabbS$ $(B \rightarrow bS)$
$\Rightarrow aabbaB$ $(S \rightarrow aB)$
$\Rightarrow aabbab$ $(B \rightarrow b)$



$S \Rightarrow aB$ $(S \rightarrow aB)$
$\Rightarrow aaBB$ $(B \rightarrow aBB)$
$\Rightarrow aabSB$ $(B \rightarrow bS)$
$\Rightarrow aabbAB$ $(S \rightarrow bA)$
$\Rightarrow aabbaB$ $(A \rightarrow a)$
$\Rightarrow aabbab$ $(B \rightarrow b)$



Given grammar is ambiguous.

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Q4. Obtain the string aaabbabbba by applying left most derivation and the parse tree for the grammar shown below. Is it possible to obtain the same string again by applying leftmost derivation but by selecting different productions?

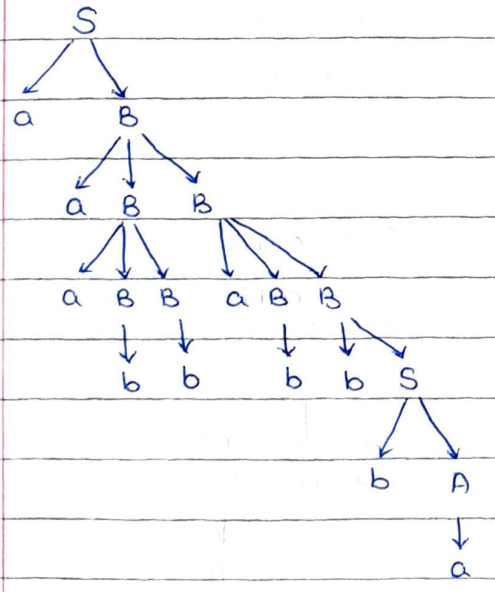
$S \rightarrow aB \mid bA$

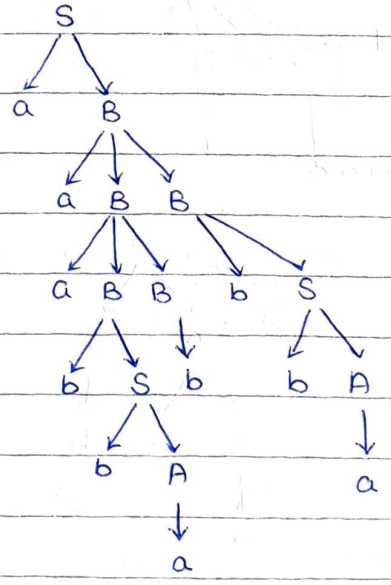
$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow bS \mid aBB \mid b$

$S \Rightarrow aB \quad (S \rightarrow aB)$
$\Rightarrow aaBB \quad (B \rightarrow aBB)$
$\Rightarrow aaaBBB \quad (B \rightarrow aBB)$
$\Rightarrow aaabBB \quad (B \rightarrow b)$
$\Rightarrow aaabbB \quad (B \rightarrow b)$
$\Rightarrow aaabbaBB \quad (B \rightarrow aBB)$
$\Rightarrow aaabbabB \quad (B \rightarrow b)$
$\Rightarrow aaabbabbS \quad (B \rightarrow bS)$
$\Rightarrow aaabbabbbA \quad (S \rightarrow bA)$
$\Rightarrow aaabbabbba \quad (A \rightarrow a)$

$S \Rightarrow aB \quad (S \rightarrow aB)$
$\Rightarrow aaBB \quad (B \rightarrow aBB)$
$\Rightarrow aaaBBB \quad (B \rightarrow aBB)$
$\Rightarrow aaabSBB \quad (B \rightarrow bS)$
$\Rightarrow aaabbABB \quad (S \rightarrow bA)$
$\Rightarrow aaabbaBB \quad (A \rightarrow a)$
$\Rightarrow aaabbabB \quad (B \rightarrow b)$
$\Rightarrow aaabbabbS \quad (B \rightarrow bS)$
$\Rightarrow aaabbabbbA \quad (S \rightarrow bA)$
$\Rightarrow aaabbabbba \quad (A \rightarrow a)$





Q5. Is the following grammar ambiguous ?

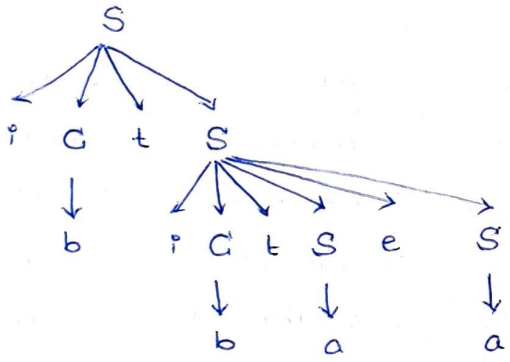
$S \rightarrow iCtS \mid iCtSeS \mid a$

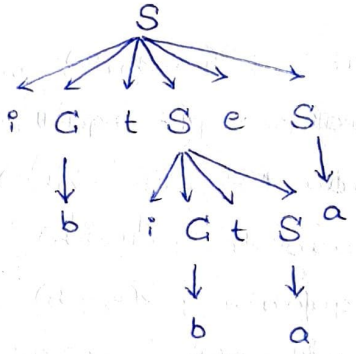
$C \rightarrow b$

String is ibtibtaea

$S \Rightarrow iCtS$
$\Rightarrow ibtS$
$\Rightarrow ibtiCtSeS$
$\Rightarrow ibtibtSeS$
$\Rightarrow ibtibtaeS$
$\Rightarrow ibtibtaea$

$S \Rightarrow iCtSeS$
$\Rightarrow ibtSeS$
$\Rightarrow ibtiCtSeS$
$\Rightarrow ibtibtSeS$
$\Rightarrow ibtibtaeS$
$\Rightarrow ibtibtaea$





Since there are two different parse trees for the string 'ibtibtaea' by applying leftmost derivation, the given grammar is ambiguous.

Q6. Is the grammar ambiguous ?

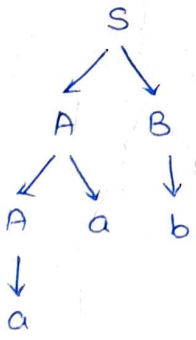
$S \rightarrow AB \mid aaB$

$A \rightarrow a \mid Aa$

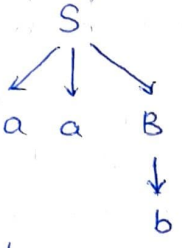
$B \rightarrow b$

String is aab

$S \Rightarrow AB$
$\Rightarrow AaB$
$\Rightarrow aaB$
$\Rightarrow aab$



$S \Rightarrow aaB$
$\Rightarrow aab$



Since there are two parse trees for the string aab, the given grammar is ambiguous.

Q7. Show that the following grammar is ambiguous

$S \rightarrow aSbS$

$S \rightarrow bSaS$

$S \rightarrow \varepsilon$

String is aababb

$S \Rightarrow aSbS$ $(S \rightarrow aSbS)$

$\Rightarrow aaSbSbS$ $(S \rightarrow aSbS)$

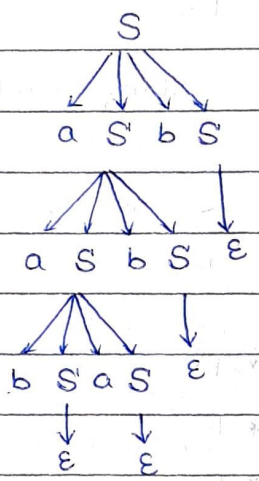
$\Rightarrow aabSaSbSbS$ $(S \rightarrow bSaS)$

$\Rightarrow aababSbSbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aababSbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aababbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aababb$ $(S \rightarrow \varepsilon)$



$S \Rightarrow aSbS$ $(S \rightarrow aSbS)$

$\Rightarrow aaSbSbS$ $(S \rightarrow aSbS)$

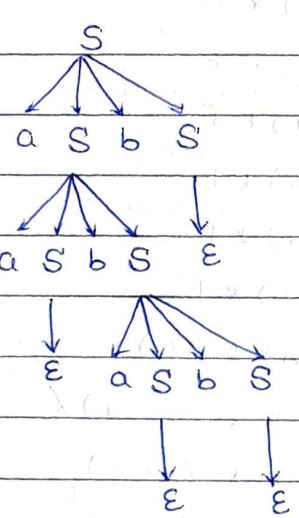
$\Rightarrow aabSbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aabaSbSbS$ $(S \rightarrow aSbS)$

$\Rightarrow aababSbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aababbS$ $(S \rightarrow \varepsilon)$

$\Rightarrow aababb$ $(S \rightarrow \varepsilon)$



Since there are two parse trees for the string aababb by applying leftmost derivation, the grammar is ambiguous.

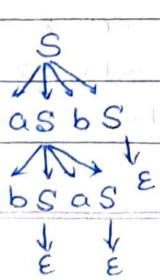
(OR)

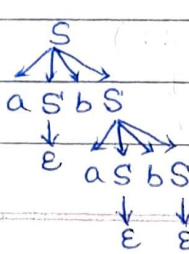
String is abab

$S \Rightarrow aSbS$

$\Rightarrow abSaSbS$

$\Rightarrow abab$





$S \Rightarrow aSbS$

$\Rightarrow abS$

$\Rightarrow abaSbS \Rightarrow abab$

Q8. Obtain the unambiguous grammar for the grammar shown below

$E \rightarrow E+E \mid E-E \mid E*E \mid E/E \mid (E) \mid I$

$I \rightarrow a \mid b \mid c$

and obtain the derivation for the expression $(a+b)*(a-b)$.

Unambiguous grammar

(4) $E \rightarrow E+T \mid E-T \mid T$

(3) $T \rightarrow T*F \mid T/F \mid F$

(2) $F \rightarrow (E) \mid I$

(1) $I \rightarrow a \mid b \mid c$

This grammar can be converted into unambiguous grammar based on the precedence.

1	a b c	left associative
2	( )	
3	* /	
4	+ -	

Derivation for the string $(a+b)*(a-b)$

$E \Rightarrow T$
$\Rightarrow T*F$
$\Rightarrow F*F$
$\Rightarrow (E)*F$
$\Rightarrow (E+T)*F$
$\Rightarrow (T+T)*F$
$\Rightarrow (F+T)*F$
$\Rightarrow (I+T)*F$
$\Rightarrow (a+T)*F$
$\Rightarrow (a+F)*F$
$\Rightarrow (a+I)*F$
$\Rightarrow (a+b)*F$
$\Rightarrow (a+b)*(E)$
$\Rightarrow (a+b)*(E-T)$
$\Rightarrow (a+b)*(T-T)$
$\Rightarrow (a+b)*(F-T)$
$\Rightarrow (a+b)*(I-T)$
$\Rightarrow (a+b)*(a-T)$
$\Rightarrow (a+b)*(a-F)$
$\Rightarrow (a+b)*(a-I)$
$\Rightarrow (a+b)*(a-b)$

NOTE:

<1> If we assume + & - are having highest priority compared to that of * & / and all the operators are left associate, the grammar will be of the form:

$E \rightarrow E*T \mid E/T \mid T$

$T \rightarrow T+F \mid T-F \mid F$

$F \rightarrow (E) \mid I$

$I \rightarrow a \mid b \mid c$

<2> If we assume + & - are having highest priority compared to that of * and / and all the operators are right associate, the grammar will be of the form:

$E \rightarrow T*E \mid T/E \mid T$

$T \rightarrow F+T \mid F-T \mid F$

$F \rightarrow (E) \mid I$

$I \rightarrow a \mid b \mid c$

<3> If we assume + & - are having highest priority & left associate when compared to that of * and / & if * and / operators are right associate, the grammar will be of the form:

$E \rightarrow T*E \mid T/E \mid T$

$T \rightarrow T+F \mid T-F \mid F$

$F \rightarrow (E) \mid I$

$I \rightarrow a \mid b \mid c$

Inherently Ambiguous Grammar

If there exists a Language L for which there is no unambiguous grammar, then the language is called inherently ambiguous language & the grammar from which the language is derived is called inherently ambiguous grammar.

Example : $L = \{a^n b^n c^m d^m \mid m \geq 1, n \geq 1\} \cup \{a^n b^m c^m d^n \mid m \geq 1, n \geq 1\}$

Grammar

$S \rightarrow AB \mid C$

$A \rightarrow aAb \mid ab$

$B \rightarrow cBd \mid cd$

$C \rightarrow aCd \mid aDd$

$D \rightarrow bDc \mid bc$

String abcd - Derivation

$S \Rightarrow AB$

$\Rightarrow abB$

$\Rightarrow abcd$

String abcd - Derivation

$S \Rightarrow C$

$\Rightarrow aDd$

$\Rightarrow abcd$

So, if we write the parse trees for both the derivations, the parse trees are different & naturally the grammar is ambiguous.

It is not possible to obtain the unambiguous grammar for this & so it is inherently ambiguous grammar.

Applications of Context Free Grammars

- Parsers
- YACC Parser
- Markup languages
- XML
- finite automaton
- Digital Design

* Finite Automata

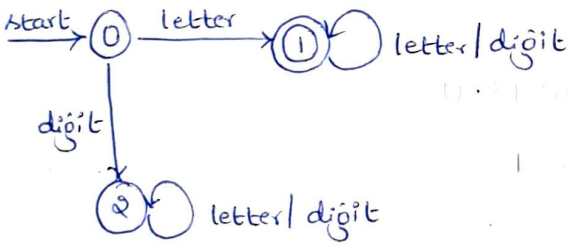
Obtain the grammar to identify an identifier.

<identifier> → <letter> <letter_digit>

<letter_digit> → <letter> <letter_digit> | <digit> <letter_digit> | ε

<letter> → a|b|...|z|A|B|...|Z

<digit> → 0|1|2|.... |9



* Digital Design

An automaton will be very useful in digital design because many of the circuits are based on concepts from aditomata theory. Even though logical implementation of a circuit is through transistors, resistors, gates, flip-flops etc., an automaton serves a bridge between its implementation and functional description of a circuit.

* YACC Parser-generator

The UNIX system provides a YACC command using which efficient parsers can be generated. The input to this command is a CFG but represented using different notations. In the notation used which is slightly different from that of notations used in CFG, each production is associated with an action which is the C code to be executed when the parse tree is created.

Example: E → E+E | E-E
E → E*E | E/E
E → (E) | I
I → a|b|c

using YACC notation can be written as,

```
E : I                {...}
  | E '+' E          {...}
  | E '-' E          {...}
  | E '*' E          {...}
  | E '/' E          {...}
  | '(' E ')'        {...}
  ;
I : 'a'              {...}
  | 'b'              {...}
  | 'c'              {...}
  ;
```

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* Markup languages

The most familiar markup language is HTML (Hyper Text Markup Language). HTML is used to create Hypertext documents for use on the World Wide Web. In HTML a block of text is surrounded with codes that indicate how it should appear on the computer screen.

* XML

XML stands for Extensible Markup Language. XML is a system for defining, validating and sharing document formats. XML uses tags to distinguish document structures, and attributes. Instead of concentrating on formatting the text, XML is used to define the semantics.

Jan 2016

Q1. Obtain a grammar to generate integers and write derivation for the unsigned integer 1965. (8 M)

$G = (V, T, P, S)$ where $V = \{D, S, N, I\}$

$T = \{+, -, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

S : I is the start state

P : $\{$ $I \rightarrow N \mid SN$

$N \rightarrow D \mid DN \mid ND$

$S \rightarrow + \mid - \mid \varepsilon$

$D \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

$\}$

Derivation for the integer 1965

$I \underset{lm}{\Rightarrow} N$

$\underset{lm}{\Rightarrow} DN$ $(N \rightarrow DN)$

$\underset{lm}{\Rightarrow} 1N$ $(D \rightarrow 1)$

$\underset{lm}{\Rightarrow} 1DN$ $(N \rightarrow DN)$

$\underset{lm}{\Rightarrow} 19N$ $(D \rightarrow 9)$

$\underset{lm}{\Rightarrow} 19DN$ $(N \rightarrow DN)$

$\underset{lm}{\Rightarrow} 196N$ $(D \rightarrow 6)$

$\underset{lm}{\Rightarrow} 196D \underset{lm}{\Rightarrow} 1965$ $(N \rightarrow D)$ $(D \rightarrow 5)$

Consider the grammar

$S \to aS \mid aSbS \mid \varepsilon$

Is above grammar ambiguous? Show that the string aab has two

i. Parse trees

ii. Left most derivations

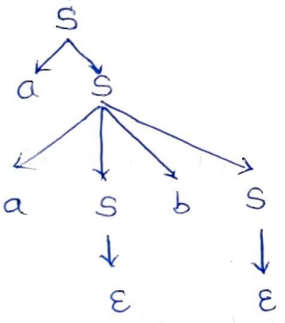
iii. Right most derivations

(12M)

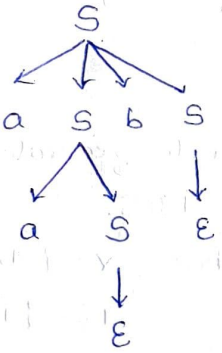
Given grammar is ambiguous.

Parse trees

$S \Rightarrow aS$
$\Rightarrow aaSbS \quad (S \to aSbS)$
$\Rightarrow aaSb \quad (S \to \varepsilon)$
$\Rightarrow aab \quad (S \to \varepsilon)$



$S \Rightarrow aSbS$
$\Rightarrow aaSbS \quad (S \to aS)$
$\Rightarrow aabS \quad (S \to \varepsilon)$
$\Rightarrow aab \quad (S \to \varepsilon)$



Left most derivations

$S \underset{lm}{\Rightarrow} aS$
$\underset{lm}{\Rightarrow} aaSbS \quad (S \to aSbS)$
$\underset{lm}{\Rightarrow} aabS \quad (S \to \varepsilon)$
$\underset{lm}{\Rightarrow} aab \quad (S \to \varepsilon)$

$S \underset{lm}{\Rightarrow} aSbS$
$\underset{lm}{\Rightarrow} aaSbS \quad (S \to aS)$
$\underset{lm}{\Rightarrow} aabS \quad (S \to \varepsilon)$
$\underset{lm}{\Rightarrow} aab \quad (S \to \varepsilon)$

Right most derivations

$S \underset{rm}{\Rightarrow} aS$
$\underset{rm}{\Rightarrow} aaSbS \quad (S \to aSbS)$
$\underset{rm}{\Rightarrow} aaSb \quad (S \to \varepsilon)$
$\underset{rm}{\Rightarrow} aab \quad (S \to \varepsilon)$

$S \underset{rm}{\Rightarrow} aSbS$
$\underset{rm}{\Rightarrow} aSb \quad (S \to \varepsilon)$
$\underset{rm}{\Rightarrow} aaSb \quad (S \to aS)$
$\underset{rm}{\Rightarrow} aab \quad (S \to \varepsilon)$

EXPERIMENT NO. ..... DATE .....

NAME OF EXPERIMENT ..... PAGE NO. 417

July 15

Q2. What is grammar? Explain the classification of grammars with examples.

Refer Notes, P.No. 401, 402. (7M)

Obtain the grammar to generate the following languages:

(i) $L = \{w \mid n_a(w) \bmod 2 = 0, \text{ where } w \in (a,b)^*\}$ P.No. 404 Q.No. 4

(ii) $L = \{w \mid w \text{ is a palindrome, where } w \in (a,b)^*\}$ P.No. 406 Q.No. 7

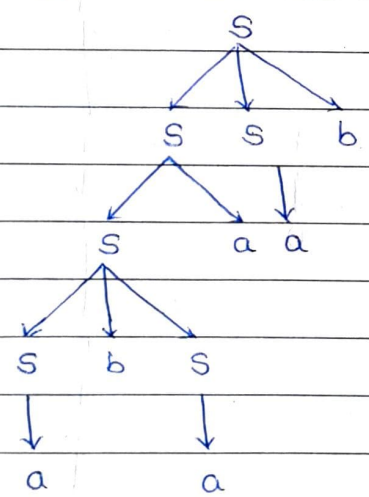
(iii) $L = \{a^n b^{2n} \mid n \geq 1\}$ $S \to abb \mid aSbb$ (6M)

Show that the following grammar is ambiguous:

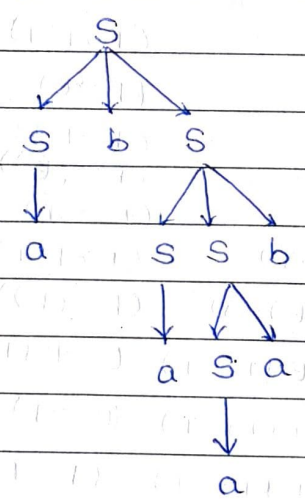
$S \to a \mid Sa \mid bSS \mid SSb \mid SbS$ (7M)

Let the string be abaaab

$S \underset{lm}{\Rightarrow} SSb$
$\underset{lm}{\Rightarrow} SaSb \quad (S \to Sa)$
$\underset{lm}{\Rightarrow} SbSaSb \quad (S \to SbS)$
$\underset{lm}{\Rightarrow} abSaSb \quad (S \to a)$
$\underset{lm}{\Rightarrow} abaaSb \quad (S \to a)$
$\underset{lm}{\Rightarrow} abaaab \quad (S \to a)$



$S \underset{lm}{\Rightarrow} SbS$
$\underset{lm}{\Rightarrow} abS \quad (S \to a)$
$\underset{lm}{\Rightarrow} abSSb \quad (S \to SSb)$
$\underset{lm}{\Rightarrow} abaSb \quad (S \to a)$
$\underset{lm}{\Rightarrow} abaSab \quad (S \to Sa)$
$\underset{lm}{\Rightarrow} abaaab \quad (S \to a)$



There are two different parse trees for the string abaaab by applying leftmost derivation. So the grammar is ambiguous.

given

Jan 15

Q3. Define a Context-Free Grammar (CFG) and also obtain the CFG's for the following languages

i. $L_1 = \{a^n w w^R b^n \mid w \in \{0,1\}^* \text{ and } n \geq 2\}$

ii. $L_2 = \{a^k b^m c^n \mid m+n = k \text{ and } m,n \geq 1\}$

iii. $L_3 = \{w \in \{a\}^* \mid |w| \bmod 3 \neq |w| \bmod 2\}$ P.No. 408, Q.No. 20 (10M)

(i) $S \rightarrow aaAbb$

$A \rightarrow aAb \mid 0A0 \mid 1A1 \mid \varepsilon$

(ii) $S \rightarrow aaSbc \mid aabc$

(iii) $S \rightarrow aa \mid aaa \mid aaaa \mid aaaaa \mid aaaaaa \mid aaaaaaS$

Consider the CFG with productions (06M)

$E \rightarrow E*T \mid T$

$T \rightarrow F-T \mid F$

$F \rightarrow (E) \mid 0 \mid 1$

Write the leftmost derivation, rightmost derivation and parse tree for the string $0-((1*0)-0)$.

$E \Rightarrow T$

$\Rightarrow F-T$ $(T \rightarrow F-T)$

$\Rightarrow 0-T$ $(F \rightarrow 0)$

$\Rightarrow 0-F$ $(T \rightarrow F)$

$\Rightarrow 0-(E)$ $(F \rightarrow (E))$

$\Rightarrow 0-(T)$ $(E \rightarrow T)$

$\Rightarrow 0-(F-T)$ $(T \rightarrow F-T)$

$\Rightarrow 0-((E)-T)$ $(F \rightarrow (E))$

$\Rightarrow 0-((E*T)-T)$ $(E \rightarrow E*T)$

$\Rightarrow 0-((T*T)-T)$ $(E \rightarrow T)$

$\Rightarrow 0-((F*T)-T)$ $(T \rightarrow F)$

$\Rightarrow 0-((1*T)-T)$ $(F \rightarrow 1)$

$\Rightarrow 0-((1*F)-T)$ $(T \rightarrow F)$

$\Rightarrow 0-((1*0)-T)$ $(F \rightarrow 0)$

$\Rightarrow 0-((1*0)-F)$ $(T \rightarrow F)$

$\Rightarrow 0-((1*0)-0)$ $(F \rightarrow 0)$

$E \Rightarrow T$

$\Rightarrow F-T$

$\Rightarrow F-F$

$\Rightarrow F-(E)$

$\Rightarrow F-(T)$

$\Rightarrow F-(F-T)$

$\Rightarrow F-(F-F)$

$\Rightarrow F-(F-0)$

$\Rightarrow F-((E)-0)$

$\Rightarrow F-((E*T)-0)$

$\Rightarrow F-((E*F)-0)$

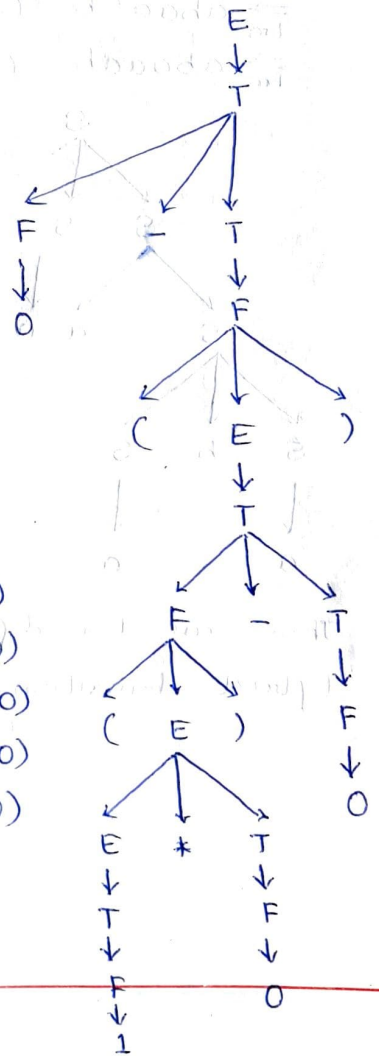
$\Rightarrow F-((E*0)-0)$

$\Rightarrow F-((T*0)-0)$

$\Rightarrow F-((F*0)-0)$

$\Rightarrow F-((1*0)-0)$

$\Rightarrow 0-((1*0)-0)$



Q4. Show that the following grammar is ambiguous. (04M)

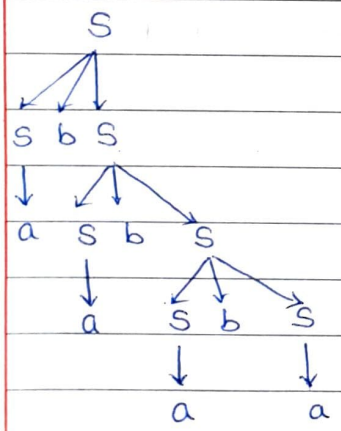
$S \rightarrow SbS$

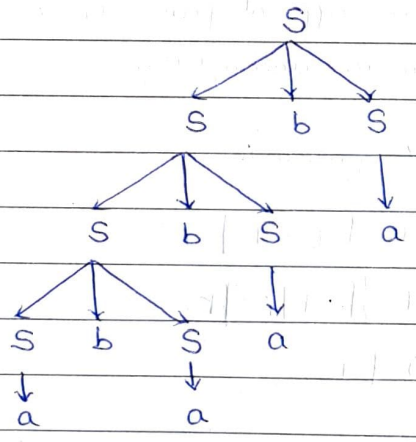
$S \rightarrow a$

Consider the string abababa

$S \underset{lm}{\Rightarrow} SbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} abS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} abSbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} ababS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} ababSbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} abababS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} abababa \quad (S \rightarrow a)$

$S \underset{lm}{\Rightarrow} SbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} SbSbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} SbSbSbS \quad (S \rightarrow SbS)$
$\underset{lm}{\Rightarrow} abSbSbS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} ababSbS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} abababS \quad (S \rightarrow a)$
$\underset{lm}{\Rightarrow} abababa \quad (S \rightarrow a)$





There are two different parse ~~differ~~ trees exists for the same string abababa by applying leftmost derivation. Hence given grammar is ambiguous.

Jul 14

Q4. Define CFG. Write CFG for the following languages. (06M)

(i) $L = \{0^n 1^n \mid n \geq 1\}$

(ii) $L = \{$ String L of a's and b's with equal number of a's and b's$\}$

(i) $S \rightarrow 0S1 \mid 01$

(ii) $S \rightarrow aSb \mid bSa \mid SS \mid \varepsilon$

Refer P.No. 402

What is an ambiguous grammar? (10M)

Show that the following grammar is ambiguous.

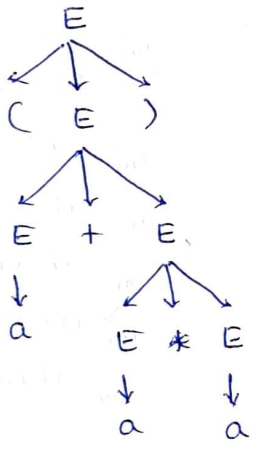
$E \rightarrow E+E \mid E-E \mid E*E \mid E/E \mid (E) \mid a$

Where E is the start symbol. find the unambiguous grammar.

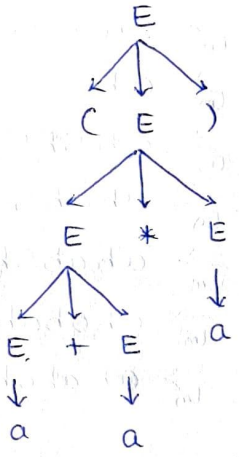
Refer P.No. 411

Consider the string (a+a*a)

$E \underset{lm}{\Rightarrow} (E)$
$\underset{lm}{\Rightarrow} (E+E)$
$\underset{lm}{\Rightarrow} (a+E)$
$\underset{lm}{\Rightarrow} (a+E*E)$
$\underset{lm}{\Rightarrow} (a+a*E)$
$\underset{lm}{\Rightarrow} (a+a*a)$



$E \underset{lm}{\Rightarrow} (E)$
$\underset{lm}{\Rightarrow} (E*E)$
$\underset{lm}{\Rightarrow} (E+E*E)$
$\underset{lm}{\Rightarrow} (a+E*E)$
$\underset{lm}{\Rightarrow} (a+a*E)$
$\underset{lm}{\Rightarrow} (a+a*a)$



There are two different parse trees exists for the string (a+a*a). Hence the given grammar is ambiguous.

Unambiguous grammar is

$E \rightarrow E+T \mid E-T \mid T$

$T \rightarrow T*F \mid T/F \mid F$

$F \rightarrow (E) \mid I$

$I \rightarrow a$

Discuss the applications of CFG. (04M)

Refer P.No. 415 and 416

Jan 14
Q5. Define CFG. Design a context free grammar for the languages: (08M)

(i) $L = \{a^i b^j c^k$, where $i = j+k$, $i, j, k \geq 0\}$

(ii) $L = \{0^{n+2} 1^n : n \geq 1\}$

Refer P. No. 402

(i) $S \rightarrow aaSbc \mid \varepsilon$

(ii) $S \rightarrow 00A$

$A \rightarrow 0A1 \mid 01$

What is an ambiguous grammar? Show that the grammar shown below is ambiguous on the string "aab". (06M)

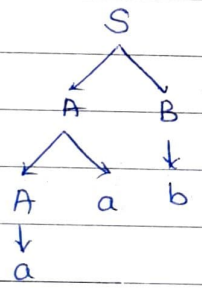
$S \rightarrow AB \mid aaB$

$A \rightarrow Aa \mid a$

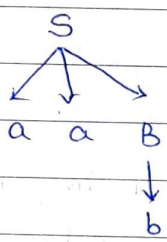
$B \rightarrow b$

Refer P. No. 411

$S \underset{lm}{\Rightarrow} AB$
$\underset{lm}{\Rightarrow} AaB$
$\underset{lm}{\Rightarrow} aaB$
$\underset{lm}{\Rightarrow} aab$



$S \underset{lm}{\Rightarrow} aaB$
$\underset{lm}{\Rightarrow} aab$

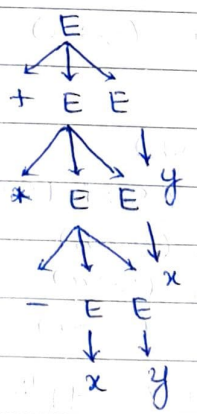


There exists two different parse trees for string aab. Hence, the given grammar is ambiguous.

Consider the grammar: $E \rightarrow +EE \mid *EE \mid -EE \mid x \mid y$. find the left most derivation, right most derivation and parse tree for the string "+*-xyxy". (06M)

$E \underset{lm}{\Rightarrow} +EE$
$\underset{lm}{\Rightarrow} +*EEE$
$\underset{lm}{\Rightarrow} +*-EEEE$
$\underset{lm}{\Rightarrow} +*-xEEE$
$\underset{lm}{\Rightarrow} +*-xyEE$
$\underset{lm}{\Rightarrow} +*-xyxE$
$\underset{lm}{\Rightarrow} +*-xyxy$

$E \underset{rm}{\Rightarrow} +EE$
$\underset{rm}{\Rightarrow} +Ey$
$\underset{rm}{\Rightarrow} +*EEy$
$\underset{rm}{\Rightarrow} +*Exy$
$\underset{rm}{\Rightarrow} +*-EExy$
$\underset{rm}{\Rightarrow} +*-Eyxy$
$\underset{rm}{\Rightarrow} +*-xyxy$



July 13

(08M)

Q6. Consider the following grammar

$E \rightarrow E+E \mid E-E$

$E \rightarrow E*E \mid E/E$

$E \rightarrow (E)$

$E \rightarrow a \mid b \mid c$

i. Obtain the left most derivation for the string (a+b*c)

ii. Obtain the right most derivation for the string (a+b)*c.

(i) $E \Rightarrow (E) \Rightarrow (E+E) \Rightarrow (a+E) \Rightarrow (a+E*E) \Rightarrow (a+b*E) \Rightarrow (a+b+c)$

(OR)

$E \Rightarrow (E) \Rightarrow (E*E) \Rightarrow (E+E*E) \Rightarrow (a+E*E) \Rightarrow (a+b*E) \Rightarrow (a+b*c)$

(ii) $E \Rightarrow E*E \Rightarrow E*c \Rightarrow (E)*c \Rightarrow (E+E)*c \Rightarrow (E+b)*c \Rightarrow (a+b)*c$

Prove that the following grammar is ambiguous, using the string "ibtibtaea".

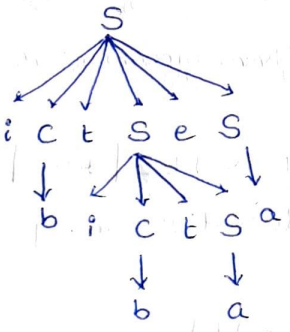
(08M)

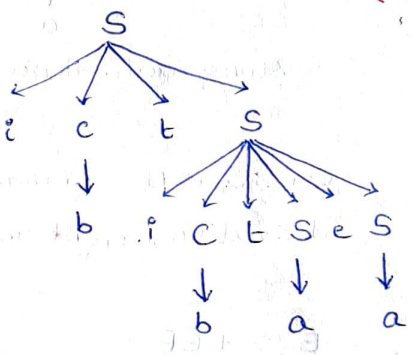
$S \rightarrow iCtS \mid iCtSeS \mid a$

$C \rightarrow b$

$S \Rightarrow iCtSeS \Rightarrow ibtSeS \Rightarrow ibtiCtSeS \Rightarrow ibtibtSeS \Rightarrow ibtibtaeS \Rightarrow ibtibtaea$

$S \Rightarrow iCtS \Rightarrow ibtS \Rightarrow ibtiCtSeS \Rightarrow ibtibtSeS \Rightarrow ibtibtaeS \Rightarrow ibtibtaea$





There exists two parse trees for the string "ibtibtaea". So, the given grammar is ambiguous.

Discuss the various applications of context free grammar. (04M)

Refer P. No. 415 and 416

Q7. Define the following terms: (05M)

i. Leftmost derivation P.No. 410
ii. Rightmost derivation P.No. 410
iii. Sentential form P.No. 405
iv. Yield of a tree P.No. 411
v. Parsing P.No. 410

Design a context free grammar for the language $L=\{ww^R : w \text{ is in } (a,b)^*\}$ (05M)

$S \rightarrow aSa \mid bSb \mid \varepsilon$

Design a context free grammar for the language $L=\{a^n b^m c^k$ where $k=m+n$, $n,m,k \geq 0\}$ (05M)

$S \rightarrow abScc \mid \varepsilon$

Show how ambiguity in grammars are verified with an example. (05M)

Def^n of ambiguous grammar.
Example with derivations and parse trees.
Refer. P. No. 411 & 412

Jun 12

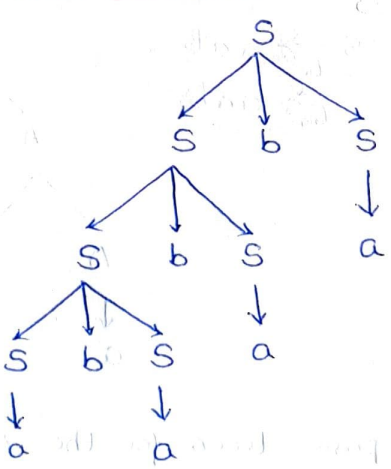
Q8. Design CFG for the following (06M)
Set of all strings of 0's and 1's, whose number of 0's equal to number of 1's.

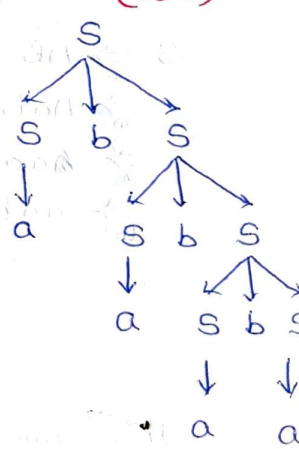
$S \rightarrow 0S1 \mid 1S0 \mid SS \mid \varepsilon$

Consider the grammar $S \rightarrow SbS \mid a$. This grammar is ambiguous. Show that particular string abababa has two i. Parse trees
ii. Left most derivations
iii. Right most derivations (10 M)

$S \Rightarrow SbS$
$\Rightarrow SbSbS$
$\Rightarrow SbSbSbS$
$\Rightarrow abSbSbS$
$\Rightarrow ababSbS$
$\Rightarrow abababS$
$\Rightarrow abababa$

$S \Rightarrow SbS$
$\Rightarrow abS$
$\Rightarrow abSbS$
$\Rightarrow ababS$
$\Rightarrow ababSbS$
$\Rightarrow abababS$
$\Rightarrow abababa$





$S \Rightarrow SbS$
$\Rightarrow SbSbS$
$\Rightarrow SbSbSbS$
$\Rightarrow SbSbSba$
$\Rightarrow SbSbaba$
$\Rightarrow Sbababa$
$\Rightarrow abababa$

$S \Rightarrow SbS$
$\Rightarrow Sba$
$\Rightarrow SbSba$
$\Rightarrow Sbaba$
$\Rightarrow SbSbaba$
$\Rightarrow Sbababa$
$\Rightarrow abababa$

Write any one application of CFG with example. (04 M)

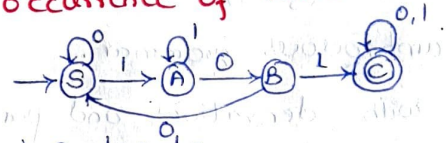
Refer P.No. 415 and 416.

Dec 11

Q9. Define context-free grammar. Obtain the CFG for the following languages (08 M)

i. $L = \{ w \mid w \in \{0,1\}^* \text{ with at least one occurrence of '101'} \}$

ii. $L = \{ a^i b^j c^k \mid i = j+k, \Sigma = \{a,b,c\} \}$



Refer P. No. 402

i. $S \rightarrow 0S \mid 1A$, $A \rightarrow 1A \mid 0B$, $B \rightarrow 1C \mid 0S$, $C \rightarrow 0C \mid 1C \mid 0 \mid \varepsilon$

ii. $S \rightarrow aaSbc \mid \varepsilon$

Explain the following with suitable examples: (06 M)

i. Left most derivation P.No.410 ii. Right most derivation P.No.410 iii. Parse tree P.No.410

What is an ambiguous grammar? Show that grammar shown below is ambiguous. (06 M)

$S \rightarrow AB \mid aaB$

$A \rightarrow Aa \mid a$

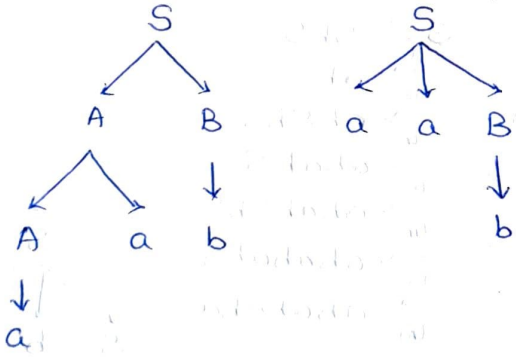
$B \rightarrow b$

Refer P. No. 411

Consider the string "aab".

$S \Rightarrow AB$
$\Rightarrow AaB$
$\Rightarrow aaB$
$\Rightarrow aab$

$S \Rightarrow aaB$
$\Rightarrow aab$



There are two parse trees for the string aab. Hence, the given grammar is ambiguous.

July 11
Q10.

Define CFG. Write a CFG for (06 M)

i. L: { Strings of a's and b's with equal number of a's and b's }

ii. $L: \{a^n b^n c^i \mid n \geq 0, i \geq 1\} \cup \{a^n b^n c^m d^m \mid n, m \geq 0\}$

ii $S_1 \rightarrow AB$
$A \rightarrow aAb \mid \varepsilon$
$B \rightarrow cB \mid c$
$S_2 \rightarrow abScd \mid \varepsilon$

$S \rightarrow S_1 \mid S_2$
$S_1 \rightarrow AB$
$S_2 \rightarrow abScd \mid \varepsilon$
$A \rightarrow aAb \mid \varepsilon$
$B \rightarrow cB \mid c$

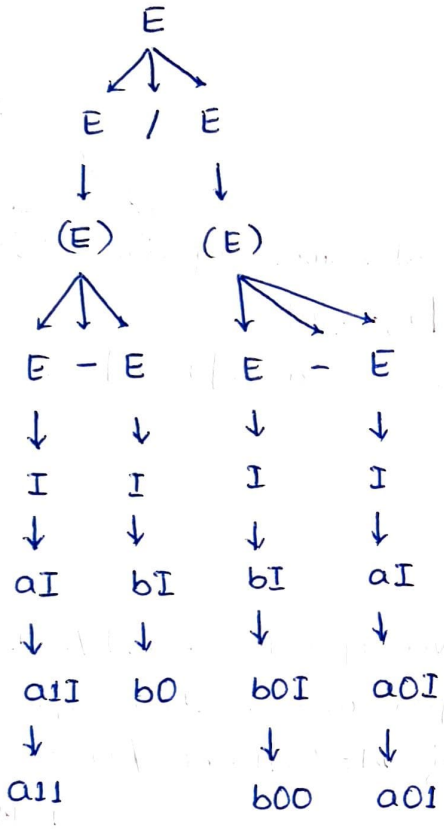
i. $S \rightarrow aSb \mid bSa \mid SS \mid \varepsilon$

Design a grammar for valid expressions over operator - and /. The arguments of expression are valid identifier over symbols a, b, 0 and 1. Derive LMD and RMD for string W: (a11 - b0) / (b00 - a01). Write parse tree for LMD. (10 M)

$E \rightarrow E-E$
$E \rightarrow E/E$
$E \rightarrow (E)$
$E \rightarrow I$
$I \rightarrow a \mid b \mid 0 \mid 1 \mid aI \mid bI \mid 0I \mid 1I$

$E \underset{LM}{\Rightarrow} E/E$
$\underset{LM}{\Rightarrow} (E)/E$
$\underset{LM}{\Rightarrow} (E-E)/E$
$\underset{LM}{\Rightarrow} (I-E)/E$
$\underset{LM}{\Rightarrow} (aI-E)/E$
$\underset{LM}{\Rightarrow} (a1I-E)/E$
$\underset{LM}{\Rightarrow} (a11-E)/E$
$\underset{LM}{\Rightarrow} (a11-I)/E$
$\underset{LM}{\Rightarrow} (a11-bI)/E$
$\underset{LM}{\Rightarrow} (a11-b0)/E$
$\underset{LM}{\Rightarrow} (a11-b0)/(E-E)$
$\underset{LM}{\Rightarrow} (a11-b0)/(I-E)$
$\underset{LM}{\Rightarrow} (a11-b0)/(bI-E)$

$\underset{LM}{\Rightarrow} (a11-b0)-(b0I-E)$
$\underset{LM}{\Rightarrow} (a11-b0)-(b00-E)$
$\underset{LM}{\Rightarrow} (a11-b0)-(b00-I)$
$\underset{LM}{\Rightarrow} (a11-b0)-(b00-aI)$
$\underset{LM}{\Rightarrow} (a11-b0)-(b00-a0I)$
$\underset{LM}{\Rightarrow} (a11-b0)-(b00-a01)$

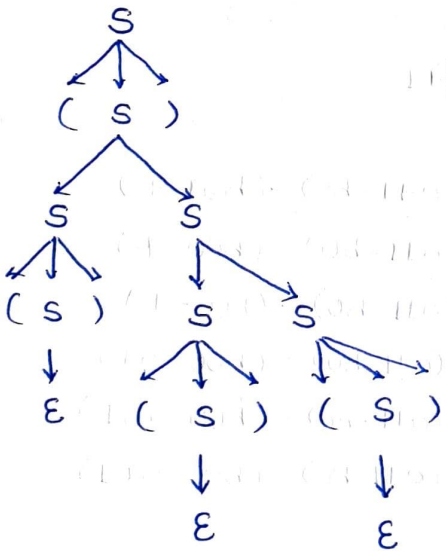


$E \Rightarrow E/E$
$\Rightarrow E/(E)$
$\Rightarrow E/(E-E)$
$\Rightarrow E/(E-I)$
$\Rightarrow E/(E-aI)$
$\Rightarrow E/(E-a0I)$
$\Rightarrow E/(E-a01)$
$\Rightarrow E/(I-a01)$
$\Rightarrow E/(bI-a01)$
$\Rightarrow E/(b0I-a01)$
$\Rightarrow E/(b00-a01)$
$\Rightarrow (E)/(b00-a01)$
$\Rightarrow (E-E)/(b00-a01)$
$\Rightarrow (E-I)/(b00-a01)$
$\Rightarrow (E-bI)/(b00-a01)$
$\Rightarrow (E-b0)/(b00-a01)$
$\Rightarrow (I-b0)/(b00-a01)$
$\Rightarrow (aI-b0)/(b00-a01)$
$\Rightarrow (a1I-b0)/(b00-a01)$
$\Rightarrow (a11-b0)/(b00-a01)$

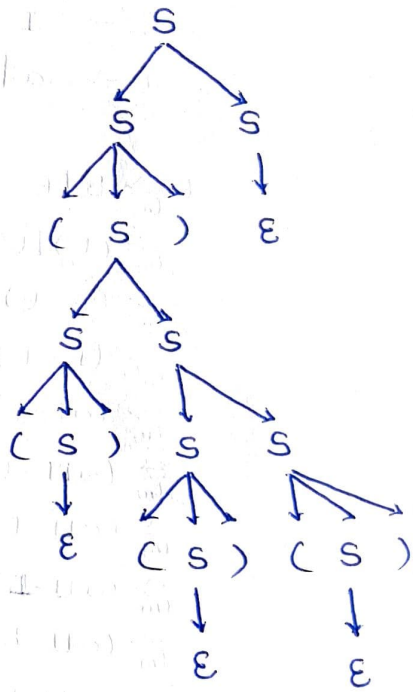
Show that the following grammar is ambiguous (04M)

$S \rightarrow SS \mid (S) \mid \varepsilon$ over $\omega = ((()()))$.

$S \Rightarrow (S)$
$\Rightarrow (SS)$
$\Rightarrow ((S)S)$
$\Rightarrow (()S)$
$\Rightarrow (()SS)$
$\Rightarrow (()(S)S)$
$\Rightarrow (()()S)$
$\Rightarrow (()()(S))$
$\Rightarrow (()()())$



$S \Rightarrow SS$
$\Rightarrow (S)S$
$\Rightarrow (SS)S$
$\Rightarrow ((S)S)S$
$\Rightarrow (()S)S$
$\Rightarrow (()SS)S$
$\Rightarrow (()(S)S)S$
$\Rightarrow (()()S)S$
$\Rightarrow (()()(S))S$
$\Rightarrow (()()())S$
$\Rightarrow (()()())$



There are two parse trees exists for the given string $((()()))$.
Hence the given grammar is ambiguous.

Dec 10
Q10

Define context free grammar. Write the grammar for the following languages: Refer P. No. 402 (07 M)

i. $L = \{0^{n+2} 1^n \mid n \geq 1\}$ $S \to 00A$, $A \to 0A1 \mid 01$

ii. $L = \{a^n b^m \mid m > n \text{ and } n \geq 0\}$ $S \to AB$, $A \to aAb \mid \varepsilon$, $B \to bB \mid b$
(OR)
$S \to aSb \mid B$, $B \to bB \mid b$

Consider the grammar G, with productions (08 M)

$S \to AbB$

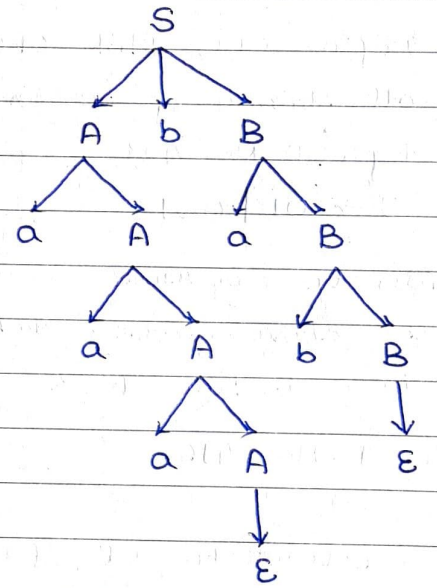
$A \to aA \mid \varepsilon$

$B \to aB \mid bB \mid \varepsilon$

Give the leftmost derivation, right most derivation and parse tree for the string aaabab.

$S \underset{lm}{\Rightarrow} AbB$
$\underset{lm}{\Rightarrow} aAbB$
$\underset{lm}{\Rightarrow} aaAbB$
$\underset{lm}{\Rightarrow} aaaAbB$
$\underset{lm}{\Rightarrow} aaabB$
$\underset{lm}{\Rightarrow} aaabaB$
$\underset{lm}{\Rightarrow} aaababB$
$\underset{lm}{\Rightarrow} aaabab$

$S \underset{rm}{\Rightarrow} AbB$
$\underset{rm}{\Rightarrow} AbaB$
$\underset{rm}{\Rightarrow} AbabB$
$\underset{rm}{\Rightarrow} Abab$
$\underset{rm}{\Rightarrow} aAbab$
$\underset{rm}{\Rightarrow} aaAbab$
$\underset{rm}{\Rightarrow} aaaAbab$
$\underset{rm}{\Rightarrow} aaabab$



What is ambiguous grammar? Show that the following grammar is ambiguous. (05 M)

$S \to AB \mid aaB$

$A \to a \mid Aa$

$B \to b$

Refer P. No. 420

June 10

Q12. Define CFG. Obtain CFG for the following languages: (10M)

i. $L = \{ww^R \mid w \in \{a,b\}^*\}$, w^R is the reversal of w

$S \to aSa \mid bSb \mid \varepsilon$

ii. $L = \{w : w$ has a substring $ab\}$

$S \to bS \mid aA$, $A \to aA \mid bB$, $B \to aB \mid bB \mid \varepsilon$

(OR)

$S \to AabA$, $A \to \varepsilon \mid aAb \mid bAa$

What is an ambiguous grammar? Show that the following grammar is ambiguous. (10M)

$E \to E+E \mid E-E \mid E*E \mid E/E \mid (E) \mid a$

where E is the start symbol. Find the unambiguous grammar.

Refer P.No. 418

Jan 10

Q13. Define CFG. Write CFG for the language $L = \{0^n1^n \mid n \geq 1\}$ i.e., the set of all strings of one or more 0's followed by an equal number of 1's. (08 M)

Refer P. No. 402.

$S \to 01 \mid 0S1$

Consider the grammar $S \to aS \mid aSbS \mid \varepsilon$

Is the above grammar ambiguous? Show in particular that the strings aab has two i. Parse trees ii. Leftmost derivation iii. Rightmost derivations. (12 M)

Refer P. No. 416

July 09

Q14 Construct the CFG for the following Languages (10 M)

i. $L = \{a^{2n}b^m \mid n \geq 0, m \geq 0\}$ $S \to AB$, $A \to \varepsilon \mid aaA$, $B \to \varepsilon \mid bB$

ii. $L = \{0^i1^j2^k \mid i=j$ or $j=k\}$ and generate Left most derivation for the string 01122.

$S \to AB$

$A \to \varepsilon \mid 0A$

$B \to \varepsilon \mid 1B2$

$S \underset{lm}{\Rightarrow} AB$

$\underset{lm}{\Rightarrow} 0AB$ $(A \to 0A)$

$\underset{lm}{\Rightarrow} 0B$ $(A \to \varepsilon)$

$\underset{lm}{\Rightarrow} 01B2$ $(B \to 1B2)$

$\underset{lm}{\Rightarrow} 011B22$ $(B \to 1B2)$

$\underset{lm}{\Rightarrow} 01122$ $(B \to \varepsilon)$

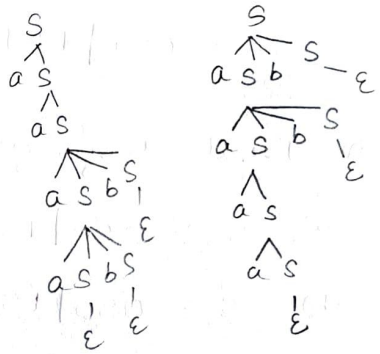
Define Ambiguous Grammar. Prove that the following grammar is Ambiguous. find an unambiguous grammar. $S \to aS \mid aSbS \mid \varepsilon$ (10 M)

Refer P.No. 411.

Refer P.No. 416.

$S \to aS \mid A$
$A \to aAbA \mid \varepsilon$ } unambiguous grammar.

String aaaabb



Jan 09

Q15. Define context-free grammar and write context free grammar for the following languages. (07 M)

i. $L = \{a^i b^j c^k \mid i+j=k,\ i>=0, j>=0\}$

ii. $L = \{a^n b^m c^k : n+2m=k\}$

i. $S \to aSc \mid A$, $A \to bAc \mid \varepsilon$

ii. $S \to aSc \mid A$, $A \to bAcc \mid \varepsilon$

Consider the grammar $E \to +EE \mid *EE \mid -EE \mid x \mid y$ (08 M)
find leftmost and rightmost derivation for the string $+*-xyxy$ and write parse tree.

Refer P.No. 419

What is ambiguous grammar? Prove that the following grammar is ambiguous on the string "aab", $S \to aS \mid aSbS \mid \varepsilon$ (05 M)

Refer P.No. 411

Refer P.No. 416

Q. Obtain a context free grammar to generate properly nested parentheses structures involving three kinds of parentheses (), [] & {}.

$S \rightarrow (S) \mid [S] \mid \{S\} \mid \varepsilon \mid SSS$

Q. What language is accepted by the following grammars ?

(a) $S \rightarrow 0S0 \mid 1S1 \mid \varepsilon$ $\quad L = \{ ww^R \mid w \in \{0,1\}^* \}$

(b) $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1$ $\quad L = \{ w \mid w \text{ is a palindrome} \}$

(c) $S \rightarrow 0S1 \mid 1S1 \mid \varepsilon$ $\quad L = \{ ww^R \mid w \in (0,1)^* \}$

$A \rightarrow 0B1 \mid 1B0$

$B \rightarrow 0B0 \mid 1B1 \mid 0 \mid 1 \mid \varepsilon$

(d) $S \rightarrow cS \mid cSdS \mid \varepsilon$ $\quad L = \{ c^i d^j \mid i \geq 1, j \geq 0 \}$

(e) $S \rightarrow cS \mid dS \mid c$ $\quad L = \{ c^i d^j \mid i \geq 1, j \geq 0 \}$

(f) $S \rightarrow SS \mid dS \mid Sd \mid c$ $\quad L = \{ c^i d^j \mid i \geq 0, j \geq 0 \}$

EXPERIMENT NO. ..... DATE .....

NAME OF EXPERIMENT ..... PAGE NO. 501

NTCSE BIT

PUSHDOWN AUTOMATA

RE
↓
RL
↓
FA

CFG
↓
CFL
↓
PDA

* Context free language can be described using context free grammars.
* FA to recognize the regular languages.
* ? automaton to recognize the context free languages.
* A DFA/NFA is not powerful enough to recognize many context free languages. (Since the FA have strict finite memories, it is not possible to construct those machines to accept the context free languages. The automaton to recognize the CFL may require additional amount of storage which will be used to store the data)

  Ex: (1) During parentheses matching

  Stack is used to remember the scanned symbols.

  (2) To count the symbols.

  $L = \{a^n b^n \mid n \geq 0\}$

* (Since DFA's or NFA's can not count and can not store the input for future reference, we are forced to have a new machine called Pushdown Automaton (PDA).)
* (PDA is a FA with the exception that PDA has an extra stack.)
* (Non-deterministic PDA, there will be ε-transitions or there may be multiple transitions from the same state on a given input when a specified symbol is there on the stacks.)
* A pushdown automata (PDA) is a seven tuple

  $M = (Q, \Sigma, T, S, q_0, Z_0, F)$

  where

  Q is set of finite states

  Σ is set of input alphabets

  T is set of stack alphabets

  S is called the transition function of M.

  Mapping from $Q \times (\Sigma \cup \varepsilon) \times T$ to finite sub set of $Q \times T^*$

  $q_0 \in Q$ is the start state of machine

  $Z_0 \in T$ is the initial symbol on the stacks

  $F \subseteq Q$ is set of final states.

tow

* Let $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ be a PDA.

The move of machine M, $\delta(p, a, Z) = (q, \alpha)$ aZ imply that, when the PDA is currently in state p, if the scanned input symbol is a and the top of the stacks is Z, the PDA enters into new state q and pushes Z on the top of the stacks. The symbols on top of the stacks and the recently pushed symbol is denoted by α. Here $p, q \in Q$, $a \in \Sigma$, $Z \in T$ & $\alpha \in T^*$.

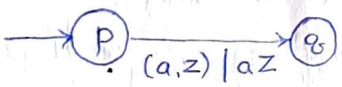
* A PDA is represented graphically using the notations as follows:
  - The states of the PDA correspond to the nodes in a graph and are represented using circles.
  - The start state of the PDA is denoted by an arrow labeled start
  - The nodes of the graph represented by two concentric circles are the final states of the PDA
  - If there is a transition of the form

    $\delta(p, a, Z) = (q, \alpha)$ aZ

    then, there will be an arc from state p to state q and the arc is labeled with $a, Z/\alpha$ indicating a is the current symbol, Z is the symbol on top of the stacks and α represent the top of the stacks along with the recently pushed symbol.

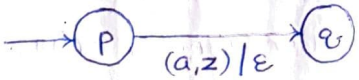
* Examples
  - $\delta(p, a, Z) = (q, aZ)$

    The PDA in current state p after scanning the input symbol $a \in \Sigma$ and if $Z \in T$ is on top of the stack, then the PDA enters into new state q pushing $a \in T$ on to the stacks.



  - $\delta(p, a, Z) = (q, \varepsilon)$

    In state p, on scanning the input symbol a, & when top of this stacks is Z, the machine enters into state q and the topmost symbol Z is deleted from the stacks.



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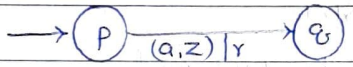
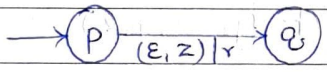
- $\delta(p, a, Z) = (q, \gamma)$

  In state p, on scanning the input symbol a and when top of the stack is Z, the machine enters into state q and topmost symbol Z on the stack is replaced by γ.



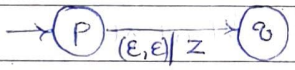
- $\delta(p, \varepsilon, Z) = (q, \gamma)$

  In state p, on scanning the empty string and when top of the stack is Z, the machine enters into q and topmost symbol Z on the stack is replaced by γ.



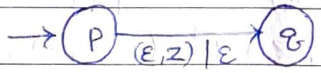
- $\delta(p, \varepsilon, \varepsilon) = (q, Z)$

  In state p, on scanning the empty string and when top of the stack is empty, the machine enters into q pushing the symbol Z on the stacks.



- $\delta(p, \varepsilon, Z) = (q, \varepsilon)$

  In state p, on scanning the empty string and when top of the stack is Z, the machine enters into q and top most symbol Z on the stack is deleted.

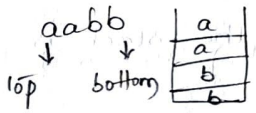


* The current configuration of PDA at any given instant can be described by an instantaneous description (ID).
* An ID gives the current state of the PDA, the remaining string to be processed and the entire contents of the stack.

* Let $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ be a PDA.

An ID is defined as 3-tuple or a triple (q, w, α) where q is the current state, w is the string to be processed & α is the current contents of stacks.

The leftmost symbol in the string α is on top of the stacks and rightmost symbol in α is at the bottom of the stacks.



* Example $(q, aw, Z\alpha)$

where, q is the current state

aw is the string to be processed

$Z\alpha$ is the current contents of the stack with Z as the topmost symbol on the stack.

* If the transition defined is $\delta(q, a, Z) = (p, \beta)$ then the new configuration obtained will be $(p, w, \beta\alpha)$

The move from the current configuration to next configuration is given by

$$(q, aw, Z\alpha) \vdash^{aZ} (p, w, \beta\alpha)$$

The configuration $(q, aw, Z\alpha)$ derives $(p, w, \beta\alpha)$ in one move.

If an arbitrary number of moves are used to move from one configuration to another configuration then the moves are denoted by the symbol $\vdash^*$ and $\vdash^+$.

* There are two cases wherein a string w is accepted by a PDA.
  - Get the final state from the start state
  - Get an empty stacks from the start state

* Let $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ be a PDA.
  - The Language $L(M)$ accepted by a final state is defined as

$$L(M) = \{ w \mid (q_0, w, Z_0) \vdash^* (p, \varepsilon, \alpha) \}$$

anything on stacks / final state / string read

for some $\alpha \in T^*$, $p \in F$ and $w \in \Sigma^*$.

  - The Language $N(M)$ accepted by an empty stack / Null stacks as

$$N(M) = \{ w \mid (q_0, w, Z_0) \vdash^* (p, \varepsilon, \varepsilon) \}$$

No final state / stacks Null / string Null

for $w \in \Sigma^*$, $q_0, p \in Q$.

[It means that when the string w is accepted by an empty stack, the final state is irrelevant, the i/p should be completely read & the stacks should be empty. Here the state p is not the final state, only thing is that the string w should be completely read & stack should be empty]

[It means that the PDA, currently in state q_0, after scanning the string w enters into a final state p. Once the machine is in state p, the input symbol should be ε & the contents of the stacks are irrelevant. Any thing can be there on the stack. The only point to remember here is that when all the symbols in string w have been read & when the machine is in the final state, the final contents of the stacks are irrelevant]

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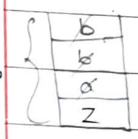
abbCbba palindrome



$q_0 \to q_1(C)$

$(q_1, \varepsilon, Z) = (q_2, Z)$

Q1. Obtain a PDA to accept the language $L(M) = \{\omega C \omega^R \mid \omega \in (a+b)^*\}$ where ω^R is reverse of ω by a final state.

The PDA to accept the language $L(M) = \{\omega C \omega^R \mid \omega \in (a+b)^*\}$ is

$M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b, C\}$

$T = \{a, b, Z_0\}$

$q_0 \in Q$ is the start state

$Z_0 \in T$ is the initial symbol on the stack

$F = \{q_2\}$ is the final state

δ is shown below

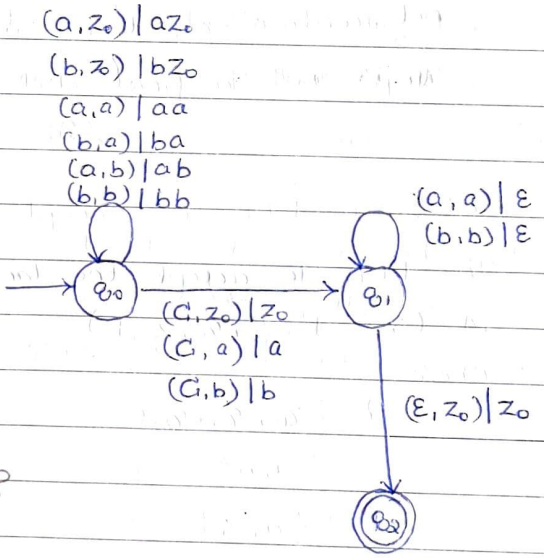
q_0:
- $\delta(q_0, a, Z_0) = (q_0, aZ_0)$
- $\delta(q_0, b, Z_0) = (q_0, bZ_0)$
- $\delta(q_0, a, a) = (q_0, aa)$
- $\delta(q_0, b, a) = (q_0, ba)$
- $\delta(q_0, a, b) = (q_0, ab)$
- $\delta(q_0, b, b) = (q_0, bb)$
- $\delta(q_0, C, Z_0) = (q_1, Z_0)$

Push "ε"

q_1:
- $\delta(q_0, C, a) = (q_1, a)$
- $\delta(q_0, C, b) = (q_1, b)$
- $\delta(q_1, a, a) = (q_1, \varepsilon)$
- $\delta(q_1, b, b) = (q_1, \varepsilon)$

pop

q_2:
- $\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$



The sequence of moves made by the PDA for the string aabCbaa is

Initial ID $(q_0, aabCbaa, Z_0) \vdash (q_0, abCbaa, aZ_0)$

$\vdash (q_0, bCbaa, aaZ_0)$

$\vdash (q_0, Cbaa, baaZ_0)$

String is accepted

$\vdash (q_1, baa, baaZ_0)$

$\vdash (q_1, aa, aaZ_0)$

$\vdash (q_1, a, aZ_0)$

$\vdash (q_1, \varepsilon, Z_0) \vdash (q_2, \varepsilon, Z_0)$ final state

The sequence of moves made by the PDA for the string aabCbab is

Initial ID $(q_0, aabCbab, Z_0) \vdash (q_0, abCbab, aZ_0)$

$\vdash (q_0, bCbab, aaZ_0)$

$\vdash (q_0, Cbab, baaZ_0)$

$\vdash (q_1, bab, baaZ_0)$

$\vdash (q_1, ab, aaZ_0)$

$\vdash (q_1, b, aZ_0)$ halts.

String is rejected.

NOTE : The same problem can be converted to accept the language by an empty stack. Only change is, instead of the final transition namely

$$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$$

replace it by transition

$$\delta(q_1, \varepsilon, Z_0) = (q_1, \varepsilon)$$

When a language is accepted by an empty stack, finally the stack should not contain any thing including Z_0.

q_1 is not a final state since there is no final state.

Q2. Obtain a PDA to accept the language $L = \{a^n b^n \mid n \geq 1\}$ by a final state.

The PDA to accept the language $L = \{a^n b^n \mid n \geq 1\}$ is given by

$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, Z_0\}$

$q_0 \in Q$

$Z_0 \in \Gamma$

$F = \{q_2\}$

δ : is shown below

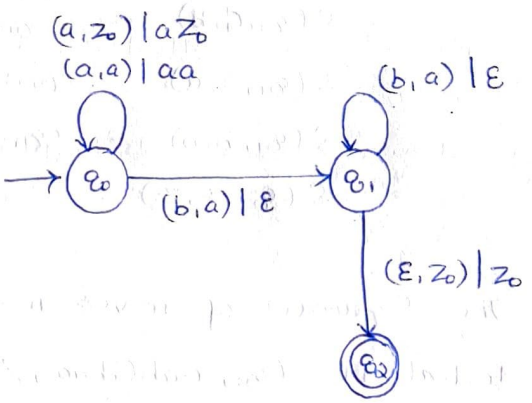
$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

$\delta(q_1, b, a) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$



The sequence of moves made by the PDA for the string aaabbb is

Initial ID $(q_0, aaabbb, Z_0) \vdash (q_0, aabbb, aZ_0)$

$\vdash (q_0, abbb, aaZ_0)$

$\vdash (q_0, bbb, aaaZ_0)$

$\vdash (q_1, bb, aaZ_0)$

$\vdash (q_1, b, aZ_0)$

$\vdash (q_1, \varepsilon, Z_0)$

$\vdash (q_2, \varepsilon, Z_0)$ final state So, the string is accepted

The sequence of moves made by the PDA for the string aabbb is

Initial ID $(q_0, aabbb, Z_0) \vdash (q_0, abbb, aZ_0)$

$\vdash (q_0, bbb, aaZ_0)$

$\vdash (q_1, bb, aZ_0)$

$\vdash (q_1, b, Z_0)$ halts, So the string is rejected by the PDA.

NOTE: By changing the final transition from

$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

to

$\delta(q_1, \varepsilon, Z_0) = (q_2, \varepsilon)$

the PDA accepted by empty stack is obtained.

q_2 is not the final state.

Q3. Obtain a PDA to accept the language $L(M) = \{w \mid w \in (a+b)^*$ and $n_a(w) = n_b(w)$ by a final state.

The PDA to accept the language $L = \{w \mid n_a(w) = n_b(w)\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1\}$ $\quad Z_0 \in T$

$\Sigma = \{a, b\}$ $\quad F = \{q_1\}$ is the final state

$T = \{a, b, Z_0\}$

$q_0 \in Q$

δ : is shown below

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, b, Z_0) = (q_0, bZ_0)$

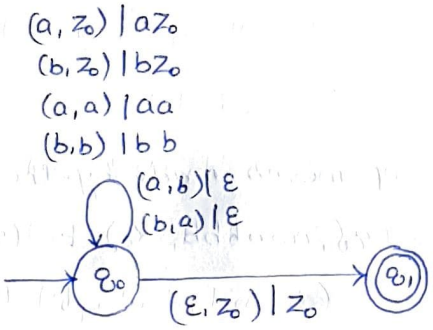
$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, \varepsilon)$

$\delta(q_0, b, a) = (q_0, \varepsilon)$

$q_1 \rightarrow \delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$



The sequence of moves made by the PDA for the string abbbaa is

Initial ID $(q_0, abbbaa, Z_0) \vdash (q_0, bbbaa, aZ_0)$

$\vdash (q_0, bbaa, Z_0)$

$\vdash (q_0, baa, bZ_0)$

$\vdash (q_0, aa, bbZ_0)$

$\vdash (q_0, a, bZ_0)$

$\vdash (q_0, \varepsilon, Z_0)$

$\vdash (q_1, \varepsilon, Z_0)$ final state. String is accepted.

The sequence of moves made by the PDA for the string aabbb is

Initial ID $(q_0, aabbb, Z_0) \vdash (q_0, abbb, aZ_0)$

$\vdash (q_0, bbb, aaZ_0)$

$\vdash (q_0, bb, aZ_0)$

$\vdash (q_0, b, Z_0)$

$\vdash (q_0, \varepsilon, bZ_0)$ Halts. String is rejected.

NOTE : To accept the language by an empty stack, change only the last transition

$$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$$

to

$$\delta(q_0, \varepsilon, Z_0) = (q_1, \varepsilon)$$

The sequence of moves made by the PDA for the string aabbab by an Empty stack is

Initial ID

$(q_0, aabbab, Z_0) \vdash (q_0, abbab, aZ_0)$

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$\vdash (q_0, bbab, aaZ_0)$

$\vdash (q_0, bab, aZ_0)$

$\vdash (q_0, ab, Z_0)$

$\vdash (q_0, b, aZ_0)$

$\vdash (q_0, \varepsilon, Z_0)$

$\vdash (q_1, \varepsilon, \varepsilon)$ final configuration. So string is accepted.

Q4. Obtain a PDA to accept a string of balanced parentheses.

Let us consider (,), [,]

The PDA to accept the language consisting of balanced parentheses is given by $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1\}$

$\Sigma = \{(,), [,]\}$

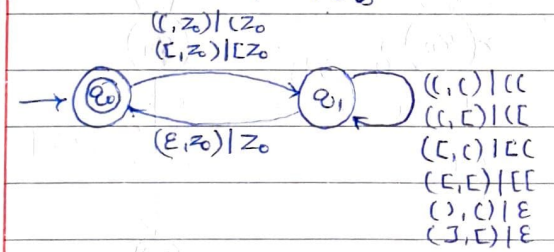
$\Gamma = \{(, [, Z_0\}$

$q_0 \in Q$

$Z_0 \in \Gamma$

$F = \{q_0\}$

Empty string is accepted



δ is shown below

$\delta(q_0, (, Z_0) = (q_1, (Z_0)$

$\delta(q_0, [, Z_0) = (q_1, [Z_0)$

$\delta(q_1, (, () = (q_1, ((\quad)$

$\delta(q_1, (, [) = (q_1, ([\quad)$

$\delta(q_1, [, () = (q_1, [(\quad)$

$\delta(q_1, [, [) = (q_1, [[\quad)$

$\delta(q_1,), () = (q_1, \varepsilon)$

$\delta(q_1,], [) = (q_1, \varepsilon)$ (ε is a valid string)

$\delta(q_1, \varepsilon, Z_0) = (q_0, Z_0)$ * accepts ε

The sequence of moves made by the PDA for the string [()()([])] is

Initial ID $(q_0, [()()([])], Z_0) \vdash (q_1, ()()([])], [Z_0)$

$\vdash (q_1,)()([])], ([Z_0)$

$\vdash (q_1, ()([])], [Z_0)$

$\vdash (q_1,)([])], ([Z_0)$

$\vdash (q_1, ([])], [Z_0)$

$\vdash (q_1, [])], ([Z_0)$

$\vdash (q_1,])], [([Z_0)$

$\vdash (q_1,)], ([Z_0)$

$\vdash (q_1,], [Z_0) \vdash (q_1, \varepsilon, Z_0) \vdash (q_0, \varepsilon, Z_0)$

String is accepted.

Q5. Obtain a PDA to accept the language $L = \{w \mid w \in (a,b)^* \text{ and } n_a(w) > n_b(w)\}$

The PDA to accept the language $L = \{w \mid n_a(w) > n_b(w)\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$

where $Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

$T = \{a, b, Z_0\}$

$q_0 \in Q$

$Z_0 \in T$

$F = \{q_1\}$

δ is shown below

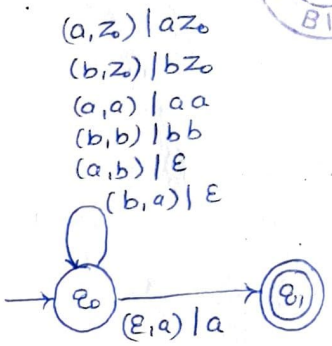
$$\begin{cases} \delta(q_0, a, Z_0) = (q_0, aZ_0) \\ \delta(q_0, b, Z_0) = (q_0, bZ_0) \end{cases}$$

$$\begin{cases} \delta(q_0, a, a) = (q_0, aa) \\ \delta(q_0, b, b) = (q_0, bb) \end{cases}$$

$$\begin{cases} \delta(q_0, a, b) = (q_0, \varepsilon) \\ \delta(q_0, b, a) = (q_0, \varepsilon) \end{cases}$$

$$\delta(q_0, \varepsilon, a) = (q_1, a) \quad *$$

$$[\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)]$$



NOTE : $L = \{w \mid w \in (a,b)^* \text{ and } n_a(w) < n_b(w)\}$

change the transition $\delta(q_0, \varepsilon, a) = (q_1, a)$ to

$\delta(q_0, \varepsilon, b) = (q_1, b)$

Q6. Obtain a PDA to accept the language $L = \{a^n b^{2n} \mid n \geq 1\}$

The PDA to accept the language $L = \{a^n b^{2n} \mid n \geq 1\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$

where $Q = \{q_0, q_1, q_2\}$

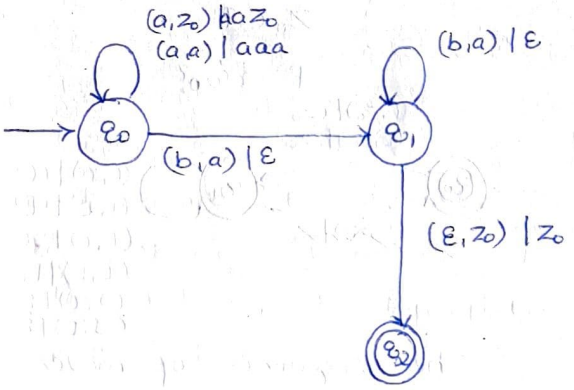
$\Sigma = \{a, b\}$

$T = \{a, Z_0\}$

$q_0 \in Q$

$Z_0 \in T$

$F = \{q_2\}$



δ is shown below

$\delta(q_0, a, Z_0) = (q_0, aaZ_0)$

$\delta(q_0, a, a) = (q_0, aaa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

$\delta(q_1, b, a) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$

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The sequence of moves made by the PDA for the string aabbbb is

Initial ID $(q_0, aabbbb, Z_0) \vdash (q_0, abbbb, aZ_0)$

$\vdash (q_0, bbbb, aaaZ_0)$

$\vdash (q_0, bbb, aaaZ_0)$

$\vdash (q_1, bb, aaZ_0)$

$\vdash (q_1, b, aZ_0)$

$\vdash (q_1, \varepsilon, Z_0)$

$\vdash (q_2, \varepsilon, Z_0)$ final configuration.

String is accepted.

The sequence of moves made by the PDA for the string aabbb is

Initial ID $(q_0, aabbb, Z_0) \vdash (q_0, abbb, aZ_0)$

$\vdash (q_0, bbb, aaaZ_0)$

$\vdash (q_0, bb, aaaZ_0)$

$\vdash (q_1, b, aaZ_0)$

$\vdash (q_1, \varepsilon, aZ_0)$ Halts.

String is rejected.

Q7. Obtain a PDA to accept the language $L = \{ww^R \mid w \in (a+b)^*\}$ by a final state.

The PDA to accept the language $L = \{ww^R \mid w \in (a,b)^*\}$ is given by

$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

where

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, b, Z_0\}$

$q_0 \in Q$

$Z_0 \in \Gamma$

$F = \{q_2\}$

δ is shown below

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$

$\delta(q_0, b, Z_0) = (q_0, bZ_0)$

$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$

① $\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, a, b) = (q_0, ab)$

② $\delta(q_0, b, b) = (q_0, bb)$

Mid point $\delta(q_0, a, a) = (q_1, \varepsilon)$

① $\delta(q_0, b, b) = (q_1, \varepsilon)$

$\delta(q_1, a, a) = (q_1, \varepsilon)$

② $\delta(q_1, b, b) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

$\delta(q_0, \varepsilon, a) = (q_1, a)$

$\delta(q_0, \varepsilon, b) = (q_1, b)$

Empty string is accepted

The sequence of moves made by the PDA for the string aabbaa is

Initial ID $(q_0, aabbaa, Z_0) \vdash (q_0, abbaa, aZ_0)$ $\vdash (q_1, bbaa, Z_0)$ Halts

$\vdash (q_0, bbaa, aaZ_0)$

$\vdash (q_0, baa, baaZ_0)$ / $\vdash (q_0, aa, bbaaZ_0)$

$\vdash (q_1, aa, aaZ_0)$ $\vdash (q_0, a, abbaaZ_0)$

$\vdash (q_1, a, aZ_0)$ $\vdash (q_1, \varepsilon, bbaaZ_0)$ / $\vdash (q_0, \varepsilon, aabbaaZ_0)$

$\vdash (q_1, \varepsilon, Z_0)$ Halts Halts

$\vdash (q_2, \varepsilon, Z_0)$ accept.

$(q_0, a, a) = (q_0, aa)$
$(q_0, a, a) = (q_1, \varepsilon)$

$(q_0, b, b) = (q_0, bb)$
$(q_0, b, b) = (q_1, \varepsilon)$

Deterministic and Non-deterministic PDA

Let $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ be a PDA. The PDA is deterministic if

* $\delta(q, a, Z)$ has only one element
* If $\delta(q, \varepsilon, Z)$ is not empty, then $\delta(q, a, Z)$ should be empty

Both the conditions should be satisfied for the PDA to be deterministic.
If one of the conditions fails, the PDA is non-deterministic.

Q1. Is the PDA to accept the language $L(M) = \{wCw^R \mid w \in (a+b)^*\}$ is deterministic?

The transitions are,

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$
$\delta(q_0, b, Z_0) = (q_0, bZ_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, a) = (q_0, ba)$
$\delta(q_0, a, b) = (q_0, ab)$
$\delta(q_0, b, b) = (q_0, bb)$
$\delta(q_0, C, Z_0) = (q_1, Z_0)$
$\delta(q_0, C, a) = (q_1, a)$
$\delta(q_0, C, b) = (q_1, b)$
$\delta(q_1, a, a) = (q_1, \varepsilon)$
$\delta(q_1, b, b) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

<1> For each $q \in Q$, $a \in \Sigma$ and $Z \in T$ there is only component defined and the first condition is satisfied.

<2> If there is an ε-transition ($\delta(q_1, \varepsilon, Z_0)$) then there should not be any transition from the state q_1 when top of the stack is Z_0, which is true.

Since the PDA satisfies both the conditions, the PDA is deterministic.

Q2. Is the PDA corresponding to the language $L = \{a^n b^n \mid n \geq 1\}$ by a final state is deterministic?

The transitions are

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, a) = (q_1, \varepsilon)$
$\delta(q_1, b, a) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$

<1> for each $q \in Q$, $a \in \Sigma$ and $Z \in \Gamma$ there exists only one definition. So, the first condition is satisfied.

<2> $\delta(q_1, \varepsilon, Z_0) = (q_2, Z_0)$ is defined. $\delta(q_1, a, Z_0)$ should not be defined which is true.

Since both the conditions are satisfied, the given PDA is deterministic.

Q3. Is the PDA to accept the language $L(M) = \{w \mid w \in (a+b)^* \text{ and } n_a(w) = n_b(w)\}$ is deterministic?

The transitions are

$\delta(q_0, a, Z_0) = (q_0, aZ_0)$
$\delta(q_0, b, Z_0) = (q_0, bZ_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, b) = (q_0, bb)$
$\delta(q_0, a, b) = (q_0, \varepsilon)$
$\delta(q_0, b, a) = (q_0, \varepsilon)$
$\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$

<1> for each $q \in Q$, $a \in \Sigma$, and $Z \in \Gamma$, there exists only one component. So, the first condition is satisfied.

<2> $\delta(q_0, \varepsilon, Z_0) = (q_1, Z_0)$ is defined. $\delta(q_0, a, Z_0)$ should not be defined. but there are two transitions.
$\delta(q_0, a, Z_0) = (q_0, aZ_0)$
$\delta(q_0, b, Z_0) = (q_0, bZ_0)$
So, the second condition is not satisfied.

The given PDA is non-deterministic PDA.

Q4. Is the PDA to accept the language consisting of balanced parentheses is deterministic?

The transitions are

$\delta(q_0, (, Z_0) = (q_1, (Z_0)$
$\delta(q_0, [, Z_0) = (q_1, [Z_0)$
$\delta(q_1, (, () = (q_1, ((), \quad \delta(q_1, [, () = (q_1, [(),$
$\delta(q_1, (, [) = (q_1, ([), \quad \delta(q_1, [, [) = (q_1, [[),$
$\delta(q_1,), () = (q_1, \varepsilon), \quad \delta(q_1,], [) = (q_1, \varepsilon),$
$\delta(q_1, \varepsilon, Z_0) = (q_0, Z_0)$

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<1> for each $q \in Q$, $a \in \Sigma$ & $Z \in T$, there exists only one component. So, the first condition is satisfied.

<2> $\delta(q_1, \varepsilon, z_0) = (q_0, z_0)$ is defined.

$\delta(q_1, a, z_0)$ should not be defined, which is true.

Since both the conditions are satisfied, the given PDA is deterministic.

Q5. Is the PDA to accept the language $L = \{ w \mid w \in (a,b)^* \text{ and } n_a(w) > n_b(w) \}$ is deterministic?

The transitions are

$\delta(q_0, a, z_0) = (q_0, aZ_0)$

$\delta(q_0, b, z_0) = (q_0, bZ_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, \varepsilon)$

$\delta(q_0, b, a) = (q_0, \varepsilon)$

$\delta(q_0, \varepsilon, a) = (q_1, a)$

<1> for each $q \in Q$, $a \in \Sigma$ and $Z \in T$, there exists only one component. So, the first condition is satisfied.

<2> $\delta(q_0, \varepsilon, a) = (q_1, a)$ is defined.

$\delta(q_0, a, a)$ should not be defined.

But there are two transitions.

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

Since second condition is not satisfied. The given PDA is non-deterministic PDA.

Q6. Is the PDA to accept the language $L = \{ a^n b^{2n} \mid n \geq 1 \}$ is deterministic?

The transitions are

$\delta(q_0, a, z_0) = (q_0, aaZ_0)$

$\delta(q_0, a, a) = (q_0, aaa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

$\delta(q_1, b, a) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, z_0) = (q_2, \varepsilon)$

(q_0, ε) for $L = \{ a^n b^{2n} \mid n \geq 0 \}$

<1> for each $q \in Q$, $a \in \Sigma$ and $Z \in T$, there exists only one definition. So, the first condition is satisfied.

<2> $\delta(q_1, \varepsilon, z_0) = (q_2, \varepsilon)$ is defined.

$\delta(q_1, a, z_0)$ should not be defined, which is true.

Since both conditions are satisfied, the given PDA is deterministic.

Q7. Is the PDA to accept the language $L = \{ ww^R \mid w \in (a+b)^* \}$ is deterministic?

The transitions are

$\delta(q_0, a, z_0) = (q_0, aZ_0)$

$\delta(q_0, b, z_0) = (q_0, bZ_0)$

$\delta(q_0, a, a) = (q_0, aa), (q_1, \varepsilon)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, b) = (q_0, bb), (q_1, \varepsilon)$

$\delta(q_1, a, a) = (q_1, \varepsilon)$

$\delta(q_1, b, b) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$

<1> There are two transitions each having two components. So, the first condition fails.

The given PDA is non-deterministic.

CFG to PDA

Greibach Normal form $\{ A \to a\alpha \quad a \in T, \ \alpha \in V^* \}$

1) Convert the grammar into GNF.
2) Let q_0 be the start state and Z_0 is the initial symbol on the stack. Without consuming any input, push the start symbol S onto the stack and change the state to q_1. The transition is $\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$
3) For each production of the form $A \to a\alpha$ introduce the transition $\delta(q_1, a, A) = (q_1, \alpha)$
4) Finally, in state q_1, without consuming any input, change the state to q_f which is an accepting state. The transition is $\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

Q1. For the below grammar $S \to aABC$
$A \to aB \mid a$
$B \to bA \mid b$
$C \to a$ **Obtain the corresponding PDA.**

Let q_0 be the start state and Z_0 is the initial symbol on the stack.

① $\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

⑤ Productions	Transitions
$S \to aABC$	$\delta(q_1, a, S) = (q_1, ABC)$
$A \to aB$	$\delta(q_1, a, A) = (q_1, B)$
$A \to a$	$\delta(q_1, a, A) = (q_1, \varepsilon)$
$B \to bA$	$\delta(q_1, b, B) = (q_1, A)$
$B \to b$	$\delta(q_1, b, B) = (q_1, \varepsilon)$
$C \to a$	$\delta(q_1, a, C) = (q_1, \varepsilon)$

③ $\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

The PDA is given by $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ where

$Q = \{q_0, q_1, q_f\}$

Terminals $\Sigma = \{a, b\}$

Non-terminals $T = \{S, A, B, C, Z_0\}$

$q_0 \in Q$

$Z_0 \in T$

$F = \{q_f\}$

δ is shown below

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$
$\delta(q_1, a, S) = (q_1, ABC)$
$\delta(q_1, a, A) = (q_1, B)$
$\delta(q_1, a, A) = (q_1, \varepsilon)$
$\delta(q_1, b, B) = (q_1, A)$
$\delta(q_1, b, B) = (q_1, \varepsilon)$
$\delta(q_1, a, C) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

Derivation for the string aabba.

$S \Rightarrow aABC$ $(S \to aABC)$
$\Rightarrow aaBBC$ $(A \to aB)$
$\Rightarrow aabBC$ $(B \to b)$
$\Rightarrow aabbC$ $(B \to b)$
$\Rightarrow aabba$ $(C \to a)$

The moves made by the PDA for the string aabba.

Initial ID

$(q_0, aabba, Z_0) \vdash (q_1, aabba, SZ_0)$
$\vdash (q_1, abba, ABCZ_0)$
$(q_1, ba, ABCZ_0) \leftarrow \vdash (q_1, bba, BBCZ_0)$
halts
$\vdash (q_1, ba, BCZ_0)$
$\vdash (q_1, a, CZ_0)$
$\vdash (q_1, \varepsilon, Z_0)$
$\vdash (q_f, \varepsilon, Z_0)$

String is accepted.

Q2. for the grammar $S \to aABB \mid aAA$
$A \to aBB \mid a$
$B \to bBB \mid A$?
$C \to a$ Obtain the corresponding PDA.

Since $B \to A$ is not in GNF. So convert the grammar to GNF.

The new grammar which is in GNF is

$S \to aABB \mid aAA$
$A \to aBB \mid a$
$B \to bBB \mid aBB \mid a$
$C \to a$

Let q_0 be the start state and Z_0 is the initial symbol on the stack.

① $\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

②

Production	Transition.
$S \to aABB$	$\delta(q_1, a, S) = (q_1, ABB)$
$S \to aAA$	$\delta(q_1, a, S) = (q_1, AA)$
$A \to aBB$	$\delta(q_1, a, A) = (q_1, BB)$
$A \to a$	$\delta(q_1, a, A) = (q_1, \varepsilon)$
$B \to bBB$	$\delta(q_1, b, B) = (q_1, BB)$
$B \to aBB$	$\delta(q_1, a, B) = (q_1, BB)$
$B \to a$	$\delta(q_1, a, B) = (q_1, \varepsilon)$
$C \to a$	$\delta(q_1, a, C) = (q_1, \varepsilon)$

③ $\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

N.I.T. CSE Jamshedpur B.I.T.

The PDA is given by $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ where

$Q = \{q_0, q_1, q_f\}$

$\Sigma = \{a, b\}$

$T = \{S, A, B, C, Z_0\}$

$q_0 \in Q$

$Z_0 \in T$

$F = \{q_f\}$

δ is shown below

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, a, S) = (q_1, ABB)$

$\delta(q_1, a, S) = (q_1, AA)$

$\delta(q_1, a, A) = (q_1, BB)$

$\delta(q_1, a, A) = (q_1, \varepsilon)$

$\delta(q_1, b, B) = (q_1, BB)$

$\delta(q_1, a, B) = (q_1, BB)$

$\delta(q_1, a, B) = (q_1, \varepsilon)$

$\delta(q_1, a, C) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

Application of GNF

Q1. Obtain a PDA to accept the language $L = \{a^n b^n \mid n \geq 1\}$

The CFG to generate the language is

$S \rightarrow aSb$

$S \rightarrow ab$

Grammar → GNF → Productions → Transitions

The corresponding grammar in GNF is

$S \rightarrow aSB$

$S \rightarrow aB$

$B \rightarrow b$

The equivalent transitions

$\delta(q_1, a, S) = (q_1, SB)$

$\delta(q_1, a, S) = (q_1, B)$

$\delta(q_1, b, B) = (q_1, \varepsilon)$

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

The PDA is given by

$M = (\{q_0, q_1, q_f\}, \{a, b\}, \{S, A, B, Z_0\}, \delta, q_0, Z_0, q_f)$

δ is shown below

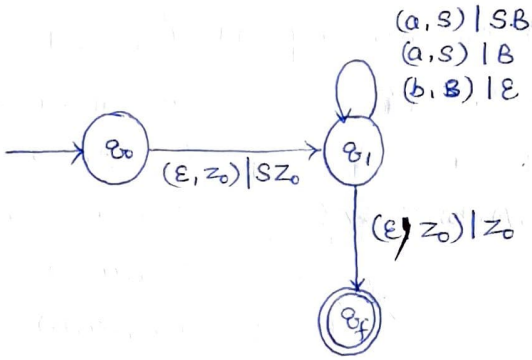
$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, a, S) = (q_1, SB)$

$\delta(q_1, a, S) = (q_1, B)$

$\delta(q_1, b, B) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$



Q2. Obtain a PDA to accept the language $L = \{ww^R : |w| \geq 1 \text{ for } w \in (a+b)^*\}$

The CFG to generate the language is

$S \rightarrow aSa \mid aa$

$S \rightarrow bSb \mid bb$

The corresponding grammar in GNF is

$S \rightarrow aSA \mid aA$

$S \rightarrow bSB \mid bB$

$A \rightarrow a$

$B \rightarrow b$

The equivalent transitions are

$\delta(q_1, a, S) = \{(q_1, SA), (q_1, A)\}$

$\delta(q_1, b, S) = \{(q_1, SB), (q_1, B)\}$

$\delta(q_1, a, A) = (q_1, \varepsilon)$

$\delta(q_1, b, B) = (q_1, \varepsilon)$

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

The PDA is given by

$M = (\{q_0, q_1, q_f\}, \{a, b\}, \{S, A, B, Z_0\}, \delta, q_0, Z_0, q_f)$

δ is shown below

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, a, S) = \{(q_1, SA), (q_1, A)\}$

$\delta(q_1, b, S) = \{(q_1, SB), (q_1, B)\}$

$\delta(q_1, a, A) = (q_1, \varepsilon)$

$\delta(q_1, b, B) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

PDA to CFG

1) The input symbols of PDA will be the terminals of CFG.

2) If the PDA moves from state q_i to state q_j on consuming the input $a \in \Sigma$ when Z is the top of the stack, then the non-terminals of CFG are the triplets of the form $(q_i Z q_j)$.

3) If q_0 is the start state and q_f is the final state then $(q_0 Z q_f)$ is the start symbol of CFG.

4) The productions of CFG can be obtained from the transitions of PDA as shown below

(a) for each transition of the form

$$\delta(q_i, a, Z) = (q_j, AB)$$

introduce the productions of the form

$$(q_i Z q_k) \rightarrow a\,(q_j A q_l)\,(q_l B q_k)$$

where q_k and q_l will take all possible values from Q.

(b) for each transition of the form

$$\delta(q_i, a, Z) = (q_j, \varepsilon)$$

introduce the production

$$(q_i Z q_j) \rightarrow a$$

Q1. Obtain a CFG for the PDA shown below $\delta(q_0, a, Z) = (q_0, AZ)$

$\delta(q_0, a, A) = (q_0, A)$

$\delta(q_0, b, A) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z) = (q_2, \varepsilon)$

In the given transitions except the second transition, all transitions are in the required form. So, convert the transition to the required form.

The corresponding transitions are

$\delta(q_0, a, A) = (q_3, \varepsilon)$ Delete 'A'

$\delta(q_3, \varepsilon, Z) = (q_0, AZ)$ insert 'A'

The new set of transitions are

$\delta(q_0, a, Z) = (q_0, AZ)$ $\quad \delta(q_0, b, A) = (q_1, \varepsilon)$

$\delta(q_0, a, A) = (q_3, \varepsilon)$ $\quad \delta(q_1, \varepsilon, Z) = (q_2, \varepsilon)$

$\delta(q_3, \varepsilon, Z) = (q_0, AZ)$

The transitions $\delta(q_0, a, A) = (q_3, \varepsilon)$

$\delta(q_0, b, A) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z) = (q_2, \varepsilon)$ can be converted into productions as shown below.

for δ of the form $\delta(q_i, a, z) = (q_j, \varepsilon)$	Resulting Productions $(q_i Z q_j) \rightarrow a$
$\delta(q_0, a, A) = (q_3, \varepsilon)$ $\delta(q_0, b, A) = (q_1, \varepsilon)$ $\delta(q_1, \varepsilon, Z) = (q_2, \varepsilon)$	$(q_0 A q_3) \rightarrow a$ $(q_0 A q_1) \rightarrow b$ $(q_1 Z q_2) \rightarrow \varepsilon$

The transitions $\delta(q_0, a, Z) = (q_0, AZ)$

$\delta(q_3, \varepsilon, Z) = (q_0, AZ)$

can be converted into productions as shown below

for δ of the form $\delta(q_i, a, z) = (q_j, AB)$	Resulting Productions $(q_i Z q_k) \rightarrow a\ (q_j A q_l)(q_l B q_k)$
$\delta(q_0, a, z) = (q_0, AZ)$	$(q_0 Z q_0) \rightarrow a(q_0 A q_0)(q_0 Z q_0) \mid a(q_0 A q_1)(q_1 Z q_0) \mid a(q_0 A q_2)(q_2 Z q_0) \mid a(q_0 A q_3)(q_3 Z q_0)$ $(q_0 Z q_1) \rightarrow a(q_0 A q_0)(q_0 Z q_1) \mid a(q_0 A q_1)(q_1 Z q_1) \mid a(q_0 A q_2)(q_2 Z q_1) \mid a(q_0 A q_3)(q_3 Z q_1)$ $(q_0 Z q_2) \rightarrow a(q_0 A q_0)(q_0 Z q_2) \mid a(q_0 A q_1)(q_1 Z q_2) \mid a(q_0 A q_2)(q_2 Z q_2) \mid a(q_0 A q_3)(q_3 Z q_2)$ $(q_0 Z q_3) \rightarrow a(q_0 A q_0)(q_0 Z q_3) \mid a(q_0 A q_1)(q_1 Z q_3) \mid a(q_0 A q_2)(q_2 Z q_3) \mid a(q_0 A q_3)(q_3 Z q_3)$

$\delta(q_3, \varepsilon, z) = (q_0, Az)$	$(q_3 z q_0) \rightarrow (q_0 A q_0)(q_0 z q_0) \mid$
	$(q_0 A q_1)(q_1 z q_0) \mid$
	$(q_0 A q_2)(q_2 z q_0) \mid$
	$(q_0 A q_3)(q_3 z q_0)$
	$(q_3 z q_1) \rightarrow (q_0 A q_0)(q_0 z q_1) \mid$
	$(q_0 A q_1)(q_1 z q_1) \mid$
	$(q_0 A q_2)(q_2 z q_1) \mid$
	$(q_0 A q_3)(q_3 z q_1)$
	$(q_3 z q_2) \rightarrow (q_0 A q_0)(q_0 z q_2) \mid$
	$(q_0 A q_1)(q_1 z q_2) \mid$
	$(q_0 A q_2)(q_2 z q_2) \mid$
	$(q_0 A q_3)(q_3 z q_2)$
	$(q_3 z q_3) \rightarrow (q_0 A q_0)(q_0 A q_3) \mid$
	$(q_0 A q_1)(q_1 z q_3) \mid$
	$(q_0 A q_2)(q_2 z q_3) \mid$
	$(q_0 A q_3)(q_3 z q_3)$

The start symbol of the grammar will be $q_0 z q_0$.

Q2. Obtain a CFG that generates the language accepted by PDA $M: (\{q_0, q_1\}, \{a, b\}, \{A, z\}, \delta, q_0, z, \{q_1\})$ with the transitions

$\delta(q_0, a, z) = (q_0, Az)$

$\delta(q_0, b, A) = (q_0, AA)$

$\delta(q_0, a, A) = (q_1, \varepsilon)$

The transition $\delta(q_0, a, A) = (q_1, \varepsilon)$ can be converted into production as shown below

For δ of the form	Resulting Productions
$\delta(q_i, a, z) = (q_j, \varepsilon)$	$(q_i z q_j) \rightarrow a$
$\delta(q_0, a, A) = (q_1, \varepsilon)$	$(q_0 A q_1) \rightarrow a$

The transitions $\delta(q_0, a, z) = (q_0, Az)$

$\delta(q_0, b, A) = (q_0, AA)$

can be converted into productions as shown below

For δ of the form $\delta(q_i, a, Z) = (q_j, AB)$	Resulting Productions $(q_i Z q_k) \rightarrow a\,(q_j A q_l)(q_l B q_k)$
$\delta(q_0, a, Z) = (q_0, AZ)$	$(q_0 Z q_0) \rightarrow a\,(q_0 A q_0)(q_0 Z q_0) \mid a\,(q_0 A q_1)(q_1 Z q_0)$
	$(q_0 Z q_1) \rightarrow a\,(q_0 A q_0)(q_0 Z q_1) \mid a\,(q_0 A q_1)(q_1 Z q_1)$
$\delta(q_0, b, A) = (q_0, AA)$	$(q_0 A q_0) \rightarrow b\,(q_0 A q_0)(q_0 A q_0) \mid b\,(q_0 A q_1)(q_1 A q_0)$
	$(q_0 A q_1) \rightarrow b\,(q_0 A q_0)(q_0 A q_1) \mid b\,(q_0 A q_1)(q_1 A q_1)$

The start symbol of the grammar will be $q_0 Z q_1$.

N T C S BIT

Q. Convert the following CFG to PDA.

i. $E \rightarrow E+E \mid E*E \mid id$.

Solution:

Convert the grammar into GNF.

$E \rightarrow id\ E_+\ E \mid id\ E_*\ E \mid id$

$E_+ \rightarrow +$

$E_* \rightarrow *$

Let q_0 be the start state and Z_0 is the initial symbol on the stack.

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

Productions	Transitions
$E \rightarrow id\ E_+\ E$	$\delta(q_1, id, E) = (q_1, E_+E)$
$E \rightarrow id\ E_*\ E$	$\delta(q_1, id, E) = (q_1, E_*E)$
$E \rightarrow id$	$\delta(q_1, id, E) = (q_1, \varepsilon)$
$E_+ \rightarrow +$	$\delta(q_1, +, E_+) = (q_1, \varepsilon)$
$E_* \rightarrow *$	$\delta(q_1, *, E_*) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

The PDA is given by $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ where

$Q = \{q_0, q_1, q_f\}$

$\Sigma = \{id, +, *\}$

$T = \{S, E, E_+, E_*, Z_0\}$

$q_0 \in Q$ is the start state.

$Z_0 \in T$ is the initial symbol on the stack

$F = \{q_f\}$ is the final state.

δ is shown below.

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, id, E) = (q_1, E_+E)$

$\delta(q_1, id, E) = (q_1, E_*E)$

$\delta(q_1, id, E) = (q_1, \varepsilon)$

$\delta(q_1, +, E_+) = (q_1, \varepsilon)$

$\delta(q_1, *, E_*) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

ii. $S \to AS|\varepsilon$

$A \to 0A1|A1|01$

Eliminate ε production

$S \to AS|A$

$A \to 0A1|A1|01$

Eliminate unit production

$S \to AS|0A1|A1|01$

$A \to 0A1|A1|01$

Convert to CNF.

① $S \to AS|A_0AA_1|AA_1|A_0A_1$

$A \to A_0AA_1|AA_1|A_0A_1$

$A_0 \to 0$

$A_1 \to 1$

② $S \to AS|A_0D_1|AA_1|A_0A_1$

$D_1 \to AA_1$

$A \to A_0D_1|AA_1|A_0A_1$

$A_0 \to 0$

$A_1 \to 1$

Let $S = A_1$, $A = A_2$, $A_0 = A_3$,

$A_1 = A_4$, $D_1 = A_5$

$A_1 \to A_2A_1|A_3A_5|A_2A_4|A_3A_4$

$A_5 \to A_2A_4$

$A_2 \to A_3A_5|A_2A_4|A_3A_4$

$A_3 \to 0$

$A_4 \to 1$

Rewrite A_2 as

$A_2 \to A_3A_4|A_3A_5|A_3A_4$

A_2 is having left recursion.

$A \to A\alpha_i \mid \beta_i$	Substitution	$A \to \beta_iA'$, $A' \to \alpha_iA' \mid \varepsilon$
$A_2 \to A_3A_4|A_3A_5|A_3A_4$	$A = A_2$ $\alpha_1 = A_4$ $\beta_1 = A_3A_5$	$A_2 \to A_3A_5A' \mid A_3A_4A'$ $A' \to A_4A' \mid A_4$
	$\beta_2 = A_3A_4$	

Replace $A' = A_6$

$A_1 \to A_2A_1 \mid A_3A_5 \mid A_2A_4 \mid A_3A_4$

$A_2 \to A_3A_5A_6 \mid A_3A_4A_6$

$A_3 \to 0$

$A_4 \to 1$

$A_5 \to A_2A_4$

$A_6 \to A_4A_6 \mid A_4$

The Grammar in GNF is,

$A_1 \to 0A_5A_6A_1 \mid 0A_4A_6A_1 \mid 0A_5 \mid 0A_5A_6A_4 \mid 0A_4A_6A_4 \mid 0A_4$

$A_2 \to 0A_5A_6 \mid 0A_4A_6$

$A_3 \to 0$

$A_4 \to 1$

$A_5 \to 0A_5A_6A_4 \mid 0A_4A_6A_4$

$A_6 \to 1A_6 \mid 1$

The PDA is given by $M = (Q, \Sigma, T, \delta, q_0, Z_0, F)$ where δ is shown below

$Q = \{q_0, q_1, q_f\}$

$\Sigma = \{0, 1\}$

$T = \{S, Z_0, A_1, A_2, A_3, A_4, A_5, A_6\}$

$q_0 \in Q$

$Z_0 \in T$

$F = \{q_f\}$

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, 0, A_1) = \{(q_1, A_5A_6A_1), (q_1, A_4A_6A_1), (q_1, A_5), (q_1, A_5A_6A_4), (q_1, A_4A_6A_4), (q_1, A_4)\}$

$\delta(q_1, 0, A_2) = \{(q_1, A_5A_6), (q_1, A_4A_6)\}$

$\delta(q_1, 0, A_3) = (q_1, \varepsilon)$

$\delta(q_1, 1, A_4) = (q_1, \varepsilon)$

$\delta(q_1, 0, A_5) = \{(q_1, A_5A_6A_4), (q_1, A_4A_6A_4)\}$

$\delta(q_1, 1, A_6) = \{(q_1, A_6), (q_1, \varepsilon)\}$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)\}$

PDA accepted by empty stack.

Let $G = (V, T, P, S)$ be a CFG. Construct the PDA P that accepts L(G) by empty stack as follows:

$P = (\{q\}, T, V \cup T, \delta, q, S)$

where transition function δ is defined by

1. for each variable A,

   $\delta(q, \varepsilon, A) = \{(q, \beta) \mid A \to \beta \text{ is a production of } P\}$

2. for each terminal a, $\delta(q, a, a) = \{(q, \varepsilon)\}$.

Q1. Convert the following grammar to PDA that accepts the same language by empty stack.

i. $E \to E+E \mid E*E \mid (E) \mid I$

$I \to Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

Variables is {E and I}

Terminals is $\{a, b, 0, 1, (,), *, +\}$

The transition function for the PDA is

i) $\delta(q, \varepsilon, I) = \{(q, a), (q, b), (q, Ia), (q, Ib), (q, I0), (q, I1)\}$

ii) $\delta(q, \varepsilon, E) = \{(q, I), (q, E+E), (q, E*E), (q, (E))\}$

iii) $\delta(q, a, a) = \{(q, \varepsilon)\}$

$\delta(q, b, b) = \{(q, \varepsilon)\}$

$\delta(q, 0, 0) = \{(q, \varepsilon)\}$

$\delta(q, 1, 1) = \{(q, \varepsilon)\}$

$\delta(q, (, () = \{(q, \varepsilon)\}$

$\delta(q,),)) = \{(q, \varepsilon)\}$

$\delta(q, +, +) = \{(q, \varepsilon)\}$

$\delta(q, *, *) = \{(q, \varepsilon)\}$

ii. $S \rightarrow 0S1 \mid A$

$A \rightarrow 1A0 \mid S \mid \varepsilon$

$V = \{S, A\}$

$T = \{0, 1\}$

a) $\delta(q, \varepsilon, S) = \{(q, 0S1), (q, A)\}$

b) $\delta(q, \varepsilon, A) = \{(q, 1A0), (q, S), (q, \varepsilon)\}$

c) $\delta(q, 0, 0) = \{(q, \varepsilon)\}$

d) $\delta(q, 1, 1) = \{(q, \varepsilon)\}$

Q. Design a PDA for the following CFG

$S \rightarrow aSb \mid bSa \mid SS \mid \varepsilon$

Grammar in GNF is,

$S \rightarrow aSB \mid bSA \mid aB \mid bA$

$B \rightarrow b$

$A \rightarrow a$

Eliminate ε productions

$S \rightarrow aSb \mid ab \mid bSa \mid ba \mid S$

Eliminate unit productions

$S \rightarrow aSb \mid bSa \mid ab \mid ba$

The PDA is given by

$M = (\{q_0, q_1, q_f\}, \{a, b\}, \{S, A, B, Z_0\}, \delta, q_0, Z_0, q_f)$

δ is shown below

$\delta(q_0, \varepsilon, Z_0) = (q_1, SZ_0)$

$\delta(q_1, a, S) = \{(q_1, SB), (q_1, B)\}$

$\delta(q_1, a, A) = (q_1, \varepsilon)$

$\delta(q_1, b, B) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_f, Z_0)$

$\delta(q_1, b, S) = \{(q_1, SA), (q_1, A)\}$